

Capstone Project
Report
on
Journal AI
“Chronicles – Write Your Journey”

Submitted By

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Acknowledgment

Our work on Journal AI; also known as “Chronicles – Write Your Journey” has reached milestones. As we moved forward in our work, we would like to thank everyone who helped us in our journey for making something great.

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We’d also like to thank every faculty who helped us review our project.

Sincerely,

Mr. Aaquif Qureshi

Mr. Asgar Datari

Mr. Malav Shah

Mr. Mrunal Shah

Abstract

Society these days is hectic as it is. People barely have time for themselves, and they don't even realize it. Kids on their phone, and adults busy with their work life, less family talks and more of internet surfing and anonymous random chat talks. This doesn't seem like a huge problem, but as it delves in, it slowly affects all human beings in the long run. This doesn't take too long to become a societal problem, and eventually affecting the whole world.

Can writing diary change your life? If you write diary daily, we can promise that you will be a different person after 21 days. I read somewhere changes takes 21 days to form a habit, and forming a habit of reviewing yourself and your actions will surely result in something better.

I remember a song named Happy Nation, and I remember a line from it:

“A man will die, but not his ideas.”

Table of Content

	Title Page	
	Certificate (From College)	I
	Acknowledgement	II
	Abstract	III
	Contents	IV
	List of Figures	VI
	List of Tables	VII
Chapter 1	Project Proposal	1
	1.1 Bigger problem	1
	1.2 Problem statement	2
Chapter 2	Literature Review	4
	2.1 Study on similar applications	4
	2.2 Comparing our features with similar applications	9
Chapter 3	Functionality Overview	10
	3.1 Overview	10
	3.2 Detailed description on features	10
Chapter 4	Project Design	13
	4.1 Overview of our project design	13
	4.2 Class diagram	14
	4.3 State diagram	15

	4.4	Use Case diagram	16
	4.5	Sequential diagram	17
	4.6	Activity diagram	23
	4.7	Data dictionary	24
	4.8	Dataflow diagram	36
	4.9	Wireframes	38
Chapter 5	Limitations		45
Chapter 6	Timeline		46
Chapter 7	Tools & Technologies		48
Chapter 8	Project Screenshots		49
Chapter 9	Conclusion & Future Work		63
	9.1	Conclusion	63
	9.2	Future Work	64
	References		65

List of Figures

Figure No.	Title	Page No.
4.2.1	Class Diagram	14
4.3.1	State Diagram	15
4.4.1	Use Case Diagram	16
4.5.1	Sequence Diagram – Login Authentication	17
4.5.2	Sequence Diagram – Signup Authentication	18
4.5.3	Sequence Diagram – View Diary and ToDo	18
4.5.4	Sequence Diagram – Edit Diary and Edit ToDo	19
4.5.5	Sequence Diagram – Delete Diary and ToDo	19
4.5.6	Sequence Diagram - Edit Settings, Profile, Add Friends, and Account Deletion	20
4.5.7	Sequence Diagram – Create Diary and ToDo	21
4.5.8	Sequence Diagram – Mark ToDo Complete	21
4.5.9	Sequence Diagram – Account Recovery	22
4.5.10	Sequence Diagram – Public Diary	22
4.6.1	Activity Diagram	23
4.8.1	DFD Level 0	36
4.8.2	DFD Level 1	37

List of Tables

Table No.	Name of Table	Page No.
4.7.1	Account Table - Data Dictionary	24
4.7.2	Login Table – Data Dictionary	25
4.7.3	Diary Table – Data Dictionary	26
4.7.4	Tag Table – Data Dictionary	27
4.7.5	Tag Entries – Data Dictionary	27
4.7.6	Notification Table – Data Dictionary	28
4.7.7	Badge Table – Data Dictionary	29
4.7.8	User Badge Info Table – Data Dictionary	30
4.7.9	Backup Table – Data Dictionary	31
4.7.10	Friend Table – Data Dictionary	32
4.7.11	Theme Table – Data Dictionary	33
4.7.12	Template Table – Data Dictionary	33
4.7.13	Reaction Table – Data Dictionary	34
4.7.14	Reaction Type Table – Data Dictionary	34
4.7.15	Streak Table – Data Dictionary	35

Chapter 1: Project Proposal

1.1 Bigger Problem

Nearly 1 in 4 adults feel lonely, 20 – 40 % of the people are introvert, most of the people spend time using social media. Fewer people can express themselves, Less and less people are taking hardships, and less and less people think about freedom, challenges. Lesser and lesser have clarity of thoughts. Most people have become more emotional, angrier, and more frustrated. People nowadays don't self-analyze themselves.

These problems are making a dire impact on individual leading to an impact on its surrounding, which leads to an impact on society, which leads to impact on whole of humanity. And most importantly it makes an impact to upcoming generations.

We believe that a small step of just keeping your daily records may change your style of living. Writing diary helps people adjust to hardships, freedom and challenges. It helps individual express intense emotions, such as anger and frustrations without harming anyone. It provides the opportunity to self-reflect and self-analyze ourselves. It helps to organize thoughts and feelings. Over a period of writing, there will be a noticeable improvement in the clarity of thoughts. It also helps in enhancing your writing skills.

Writing a diary at the end of the day not only helps to speculate the day but it helps to feel the past with an enjoyable smile and happiness. Sometimes writing a sad day today can make you smile or even laugh in the future. It can make people see their mistakes and implement them.

There is no one solution for the problem, but there is no simple solution also for this. So, it's always best to implement many solutions and let individual dragged by the most appeals ones for individuals. Small solutions may impact less but they surely do. Small impacts make big Impacts in life.

We implement Journal AI, a diary writing app named Chronicles – Write Your Journey, which makes it easy and portable to write your diary digitally.

1.2 Problem Statement

While the broader issue may be what we previously discussed, there are also specific challenges associated with maintaining a physical diary. Writing by hand can feel tedious, and consistently keeping a diary over many years can become disorganized and difficult to manage. Additionally, ensuring privacy with a physical diary is often a concern.

There is a big generation change happening, people are spending more time on digital books than on physical books. Individuals of 20th generations are spending more time on computers than anything else.

This leads to digital books, digital articles, digital newspaper and digital everything.

We create a free cross platform portable software which lets Individual write their daily dairy digitally and we tend to make it fun by adding features like To-Do List, sharing your diary, Text Summarization, our AI reacts to diary and many more.

1.2.1 Our Aim

- Involve more individuals into daily writing habits.
- Helping individual become happier, thoughtful and stress-free.
- Helping more individuals introspect themselves.
- Helping individuals get clarity of thoughts.
- Solving the bigger problem.

1.2.2 Objectives for our Application

- To keep the UI interactions as simple as possible and easy to understand.
- To keep it free to use.
- To ensure cross-platform compatibility (Android, iOS)
- To make it Secure by securely storing the user data within the device. And storing all user data on a secure cloud platform.
- To maintain user's privacy and store user information as per user consent.

1.2.3 Features in our Application

- Easy to use and easy to write Diaries
- To-Do List
- Speech to Text
- Text to Speech
- Shareable Diary – Diaries can be made public
- Manage friends
- Text Summarization
- Reminders, pop-ups, notifications to maintain Streak
- Streak
- Badges
- Digital to pdf, e-pub, book

* Kindly refer to the detailed feature description in **Chapter 3**

Chapter 2: Literature Review

2.1 Study of Similar Applications

Study of all the applications is subjected during and till the last update on September 2024, any updates afterwards are not noted down in this case study.

We may not have noted all of the applications in this case study so kindly refer to original documentations for more clarity on the respected applications.

We did case study on:

1. Notion
2. Evernotes
3. Day One
4. Journey

Purpose of the case study is to know what features can benefits our dairy writing application for better user usage.

2.1.1 Notion

Notion is a productivity and note-taking cross-platform application developed by Notion Labs, Inc. It is an online-only organizational tool on many different operating systems, with options for both free and paid subscriptions. It is based in San Francisco, California, United States.

Notion offers a smooth, simple and flexible approach in writing.

Features:

- Reordering text or image at any place in page, we can write something on text editor and can drag and drop the image or text at any place.
- Kanban Board
- Calendar, with location and Todo list integrated.
- Different templates for different works like task template to manage task or projects. Todo list for maintaining Todo, its user customizable.
- Text Summarizer, its paid.
- User Customizable templates, users can forge their own templates.
- Easy to write text editor.

Cons:

- Overly complex
- Very limited offline functions

2.1.2 Evernotes

It is a note-taking platform that helps professionals and organizations to manage tasks and share information as it is a cross-platform application and keeps devices in sync. It offers free plan with basic features, and a premium version for additional features.

Features:

- It has a fast search bar equipped with optical character recognition (OCR) which enables it to search even handwritten text in images.
- As it is available on multiple platform user can sync their notes across devices
- It can keep multiple user(teams) onboard which makes data sharing
- Users can assign tasks to other users and can track the progress
- It has AI features like text summarization, translate and can write content
- User can customize the view page for easy access
- It can link notes, meeting and reminders with calendar

Cons:

- Major number of features are not included in free plan
- Offline mode is only available on desktop
- User can create only limited number of workspaces
- Free plan has a maximum of fifty notes and one notebook per account limitation
- User Interface is confusing and feels cluttered
- Evernote's subscription prices are at higher side
- Limitations in free plans

2.1.3 Journey

Journey is a diary app that helps you capture and organize your thoughts, memories, and experiences. Available on multiple platforms, including Windows, it allows you to add text, photos, videos, and audio to your entries.

Features:

- Sync your journal to the cloud and access it on any device.
- Write journal with rich text formatting.
- View entries in timeline, calendar, map and photo view.
- Share your journal and web links with others.
- End-to-End encryption and lock your journal with passcode, fingerprint, or Face ID
- Automatically add location and weather to entries.
- Add daily reminder.
- Personalize with custom wallpapers and colored themes.
- Export entries to Microsoft Word, ePub and print to PDF.
- Daily coach programs are journaling guide to help you get started on journaling or to help you through your on-going reflections.
- Provides stories feature which includes throwback functionality.

Cons:

- There are no tutorials on how to use most of the features.
- The free function is good but most of the advanced features required subscription.
- Can't even use basic features like Heading, bullet points, font color or different font without subscription.
- Not very user friendly
- Should provide users with free trial feature for limited time
- It doesn't have autocorrect feature
- Doesn't have STT and TTS for Windows and Mac

2.1.4 Day One

Day One is a personal journaling app available for Android, macOS and iOS devices. Some features include: data syncing with multiple devices; end-to-end encryption, Markdown authoring of entries; location, weather, date, time, and other automatic metadata; quick entry menu bar; and reminders.

Features:

- Login: Login interface contains login using google, Facebook, the app's own login, or guest login.
- App User Interface: Option to create a new journal, when clicked, shows option to select cover photo from gallery, take cover photo, export (in the form of zip file or pdf file), or journal settings, which contains.
- Journal settings: Journal name, Description, Color, Stats (entries, photos), End-to-End encryption on or off, advanced settings which show all entries, show on this day, include streaks, and conceal contain. Additional option contains to auto add location or not.
- List: List which helps in making timelines, suggesting daily prompts, habit tracker, etc.
- Calendar: A day-to-day calendar which contains options for daily entries, etc.
- Media show: Media appears here when it is added to journal.
- Map: Shows a timeline of our location where we went, and we can pin it on whatever location we want to. Also shows weather information.
- Cross platform: iOS, Mac and Android.
- Sync: iCloud and Dropbox

Pricing:

- Free version: limited entries and photo storage
- Premium version: unlimited entries and photo storage

Pros:

- Multi-Platform
- Photos and audio attachment
- Tags for filtering journals
- Auto-Location tracking
- End-to-End encryption (premium feature)
- No ads

Cons:

- Many features locked behind subscription
- Complex UI
- Limited offline functionality
- Location to be entered manually.

Point of view:

- It fulfills basically every need a journal user can want, but it has a subscription model, no web version, and users cannot create templates on themselves

Areas of improvement:

- User should be able to make his own template
- End-to-End encryption should be available to all
- There should be a free premium trial
- Other options for login (Facebook, guest, etc.)
- Notifications for saving your streak for journaling

2.2 Comparing our Features with similar applications

A comparison on the features across different platforms including Chronicles.

APP FEATURES	 Chronicles	 Journey	 Notion	 Evernotes	 One Day
Diary Writing	✓	✓	*	*	✓
Custom Templates	✓	\$	✓	✓	\$
Text Summarization	✓	\$	\$	\$	
Export To Pdf,epub	✓	✓	*	*	
Speech To Text/ Text To Speech	✓	*			
Open Diary	✓				

We provide a variety of features like export to epub/pdf, custom templates, STT/TTS, summarization, all for free, while other apps don't, or have a subscription regarding that feature. Also, open diaries and add friend is a feature which is our app-exclusive, other diary writing apps don't provide it.

A lot of side features like to-do list and daily reminders are included as well.

Chapter 3: **Functionality Overview**

3.1 Overview:

This chapter provides a detailed exploration of the features designed to enhance the diary-writing experience within the app. The app integrates various tools aimed at making diary writing easy, efficient, and personalized, while also offering features that allow users to share and interact with others. These tools are intended to streamline the writing process, enhance privacy, and support user engagement through streaks, reminders, and customizable templates. Key features like speech-to-text, public sharing options, and text summarization aim to provide a comprehensive and versatile platform for users to capture and reflect on their lives.

3.2 Detailed Description on Features:

3.2.1 General Features

- We would have light/dark theme for whole app.
- We would have public/private profile instances, where users can choose to make their selected diary public or keep it private on their device itself, without storing it on internet. If they choose to be public for one instance, their profile will be visible to all their friends.
- Search and tagging, search anywhere in your diary and tag to make every diary and every page more organized.
- Backup and restore, we make it private until you said to make it public, so to make it more private and more secure, take backup of your app with a zip converted and store it anywhere else, we also provide restoration of zip which can be done easily.
- Multi User support helps one device contain more than one user instance.
- Login system with email supports (google login, mail login). Once login done, we would ask for 4 digits pin every time user wants to login.
- One click create diary feature helps user get into writing diary instantly.
- Friend list allows you and your friends to track each other's publicized diaries. {private diary stays hidden on user's device; backup is only way to safe guard the loss}.
- We would make an achievement section, where user is given badges based on criteria set.

3.2.2 Easy to use and easy to write Diaries (Text Editor)

- We aim to create a text editor which has reordering texts and/or paragraph. We would make it easy to add images in between text so users can make a complete digital diary. This is a markdown supported app so we would have all editing features of markdown. While it seems complex, Design will be simple with easy to access features while writing diary.

3.2.3 Todo List

- We do believe that To-do list is important for a diary, and for daily life. It would be a simple custom template with checkbox and list of chores, which would take its place in the most visible page of the app, first page.

3.2.4 Speech to Text

- Writing dairy on mobile, tablet or any other small device would be hard. We bring you the feature of Speech to Text so you can speak your day out, and we help you convert it into text. This would increase your productive time and also motivates you to maintain your daily dairy.

3.2.5 Text to Speech

- You want to listen what you did 5 years ago? Use Text to Speech feature. This feature allows you to get your diary converted into beautiful voice of a robot, telling you your own chronicles. User can listen to particular instance in a diary or whole of diary. Pause will be available to take a break while listening to your chronicles or someone else's chronicles.

3.2.6 Open Diary – Diaries can be made public

- This is the feature we personally believe would help individual share the view of the day from their eyes to the world. Every diary written would have its own important distinct number. You can make a part of your diary public and let the world know an instance of your life while remaining anonymous for life. Reaction will be part of expressing emotions to others chronicles.

3.2.7 Text Summarization

- A feature that would summarize your diary and make you remember what you did 5 years ago in a small snapshot. We will be using an NLP API for this one.

3.2.8 Reminders, popups, notifications to maintain Streak

- We would have reminders in the form of popups at the time you decided to write your diary or maybe if you are forgetting some habits to fulfill or maybe forgot to complete Todo List. Because we believe that consistency is the key to success and more you remember your day, the more you can figure out the mistakes.

3.2.9 Streak

- Streak is a value that increases when you write your diary or do activity in the app for a contiguous day. One day missed is streak to 0. To make it more engaging we would reward badges based on streak.

3.2.10 Digital to pdf, e-pub, book

- Everyone wants a hardcopy of their diary before an apocalypse comes, we got you cover for this also. We would convert the markdown format into a pdf, custom pdfs. We would also convert markdown into epub for book like feeling. You can also print your pdf to get physical diary feeling.

3.2.11 Templates to start writing, make templates and share them.

- We can't make templates for everyone's occasion, but you can. Share it with world and make everyone's life easy. Here we would have public templates for everyone to take a peek and copy the templates as they want.

Chapter 4: Project Design

4.1 Overview of Project Design

This chapter focuses on decomposing a project into its various components and designing each part to visualize the complete structure and gain insights into how the project will appear when fully developed. The following diagrams and tools are utilized for this purpose:

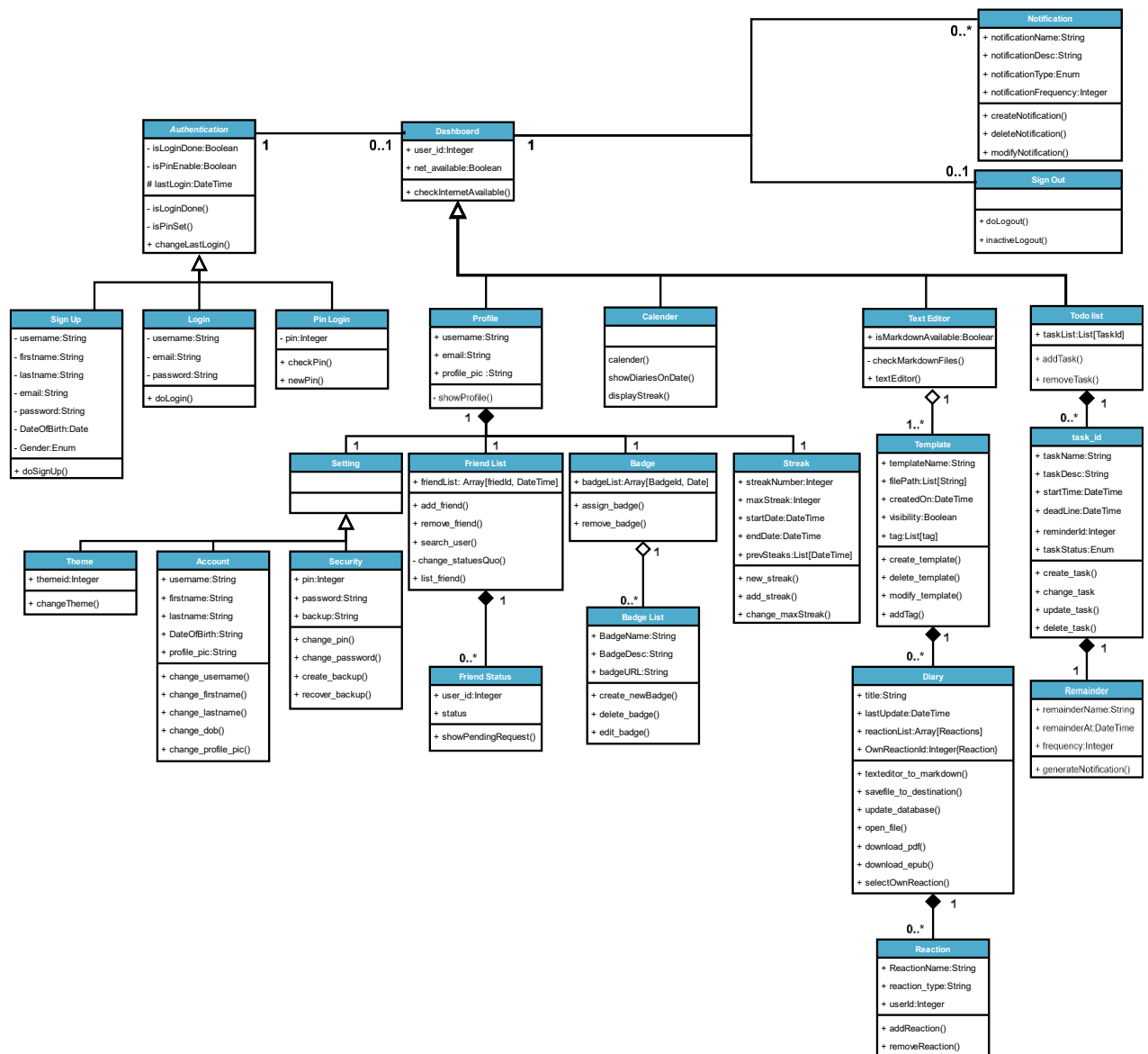
1. Class Diagram: Illustrates the classes, their attributes, methods, and the relationships between them.
2. State Diagram: Depicts the states of an object and the transitions between these states based on certain events.
3. Use Case Diagram: Shows the interactions between users (actors) and the system to achieve a goal.
4. Sequence Diagram: Visualizes the sequence of messages exchanged between objects in a particular scenario or process.
5. Activity Diagram: Represents the flow of control or activities in the system, typically used to describe the workflow of a process.
6. Data Dictionary: A detailed description of data elements, including their definitions, types, and relationships.
7. Data Flow Diagram (DFD):
 - a. Level 0: Provides a high-level overview of the system and its interactions with external entities.
 - b. Level 1: Breaks down the system into more detailed sub-processes, showing how data moves within the system.
8. Wireframes: Visual representations of the user interface (UI) layout, illustrating how the system will look from the user's perspective.

These tools are essential for understanding the system's architecture, interactions, and user interfaces, helping to build a clear vision of the final project.

4.2 Class Diagram

A UML class diagram is a visual tool that represents the structure of a system by showing its classes, attributes, methods, and the relationships between them. It helps everyone involved in a project—like developers and designers—understand how the system is organized and how its components interact.

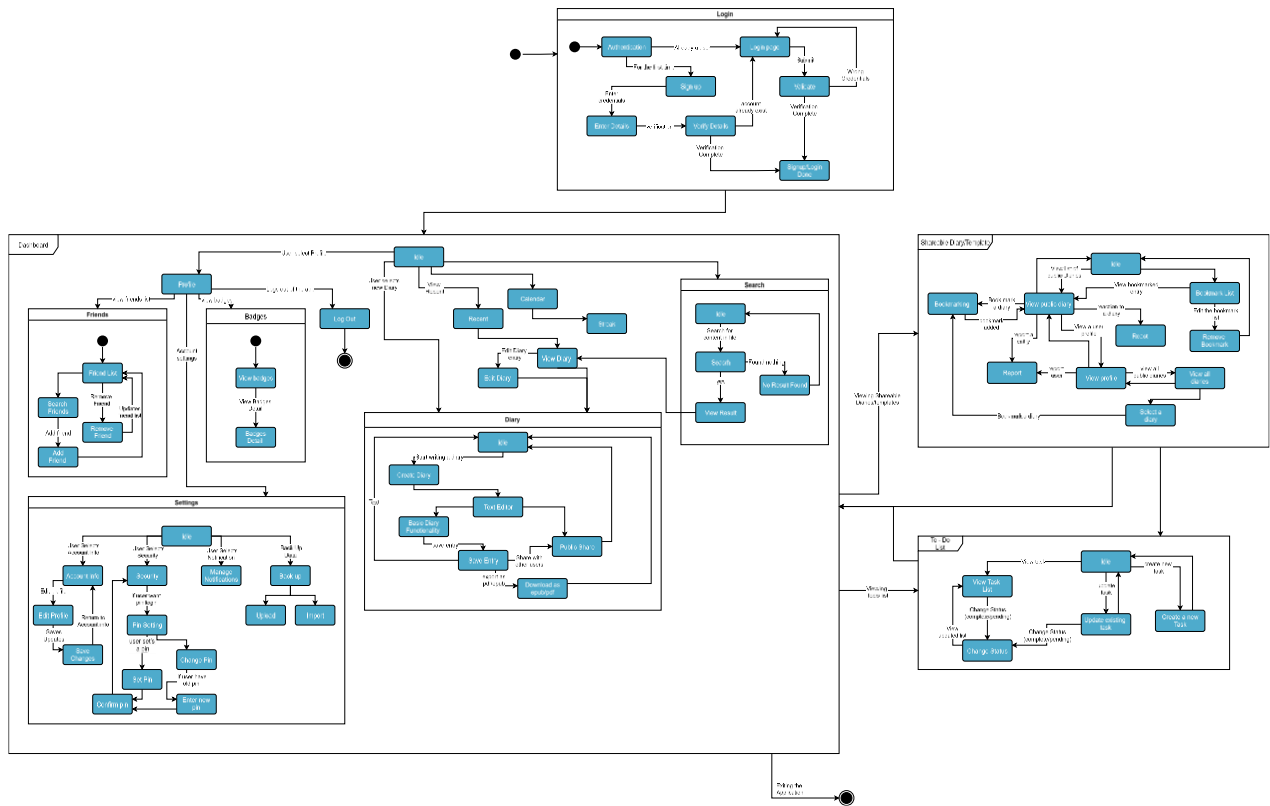
Below is our class Diagram.



4.2.1 Class Diagram

4.3 State Diagram

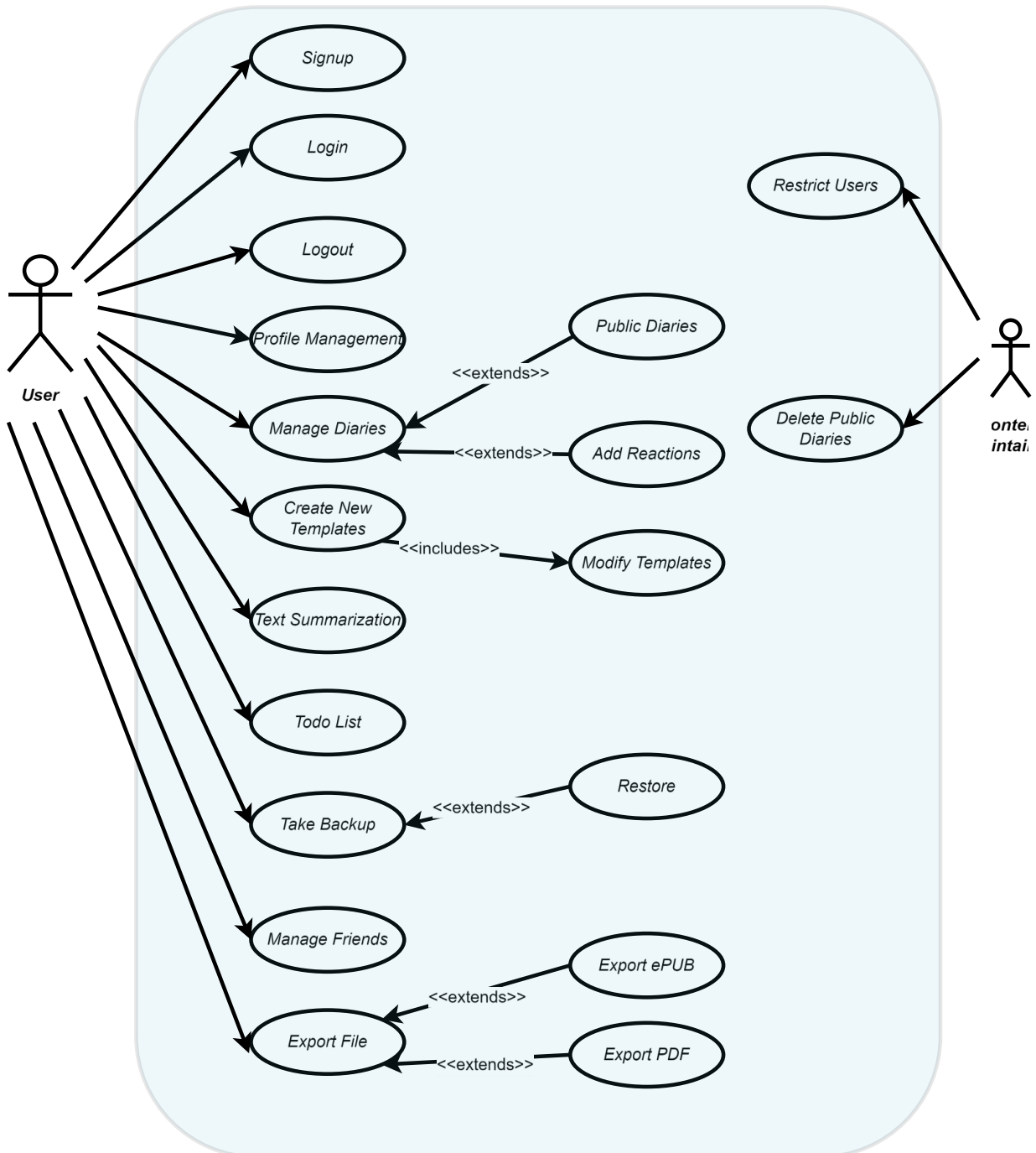
state diagram shows the behavior of classes in response to external stimuli.



4.3.1 State Diagram

4.4 Use Case Diagram

Use cases are a technique for capturing, modeling, and specifying the requirements of a system. A use case corresponds to a set of behaviors that the system may perform in interaction with its actors, and which produces an observable result that contributes to its goals.

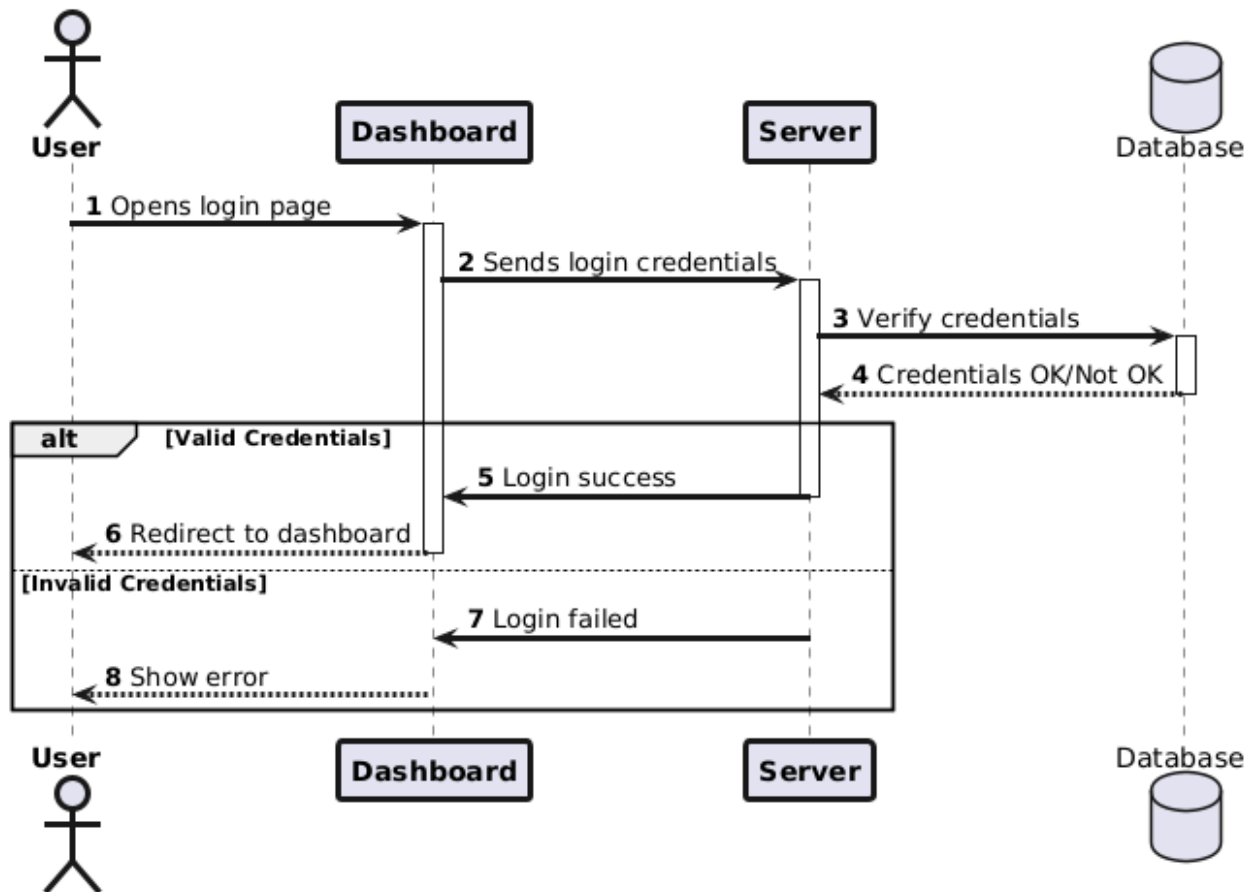


4.4.1 Use Case Diagram

4.5 Sequence Diagram

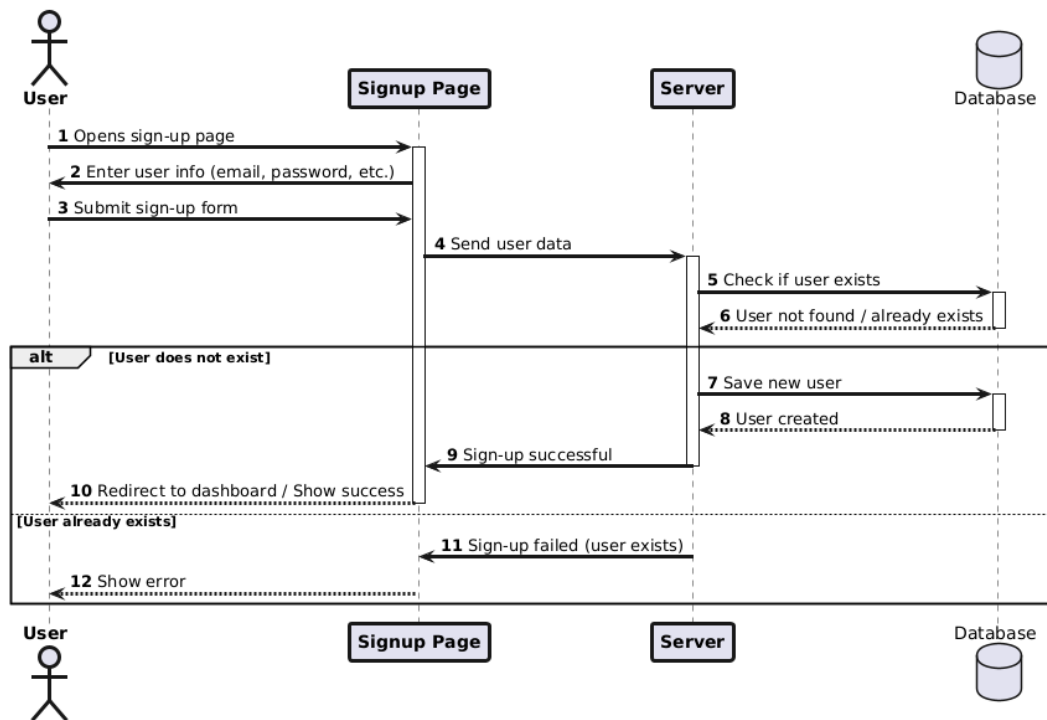
Sequence Diagram is an interaction diagram that details how operations are carried out -- what messages are sent and when.

i. Login Authentication



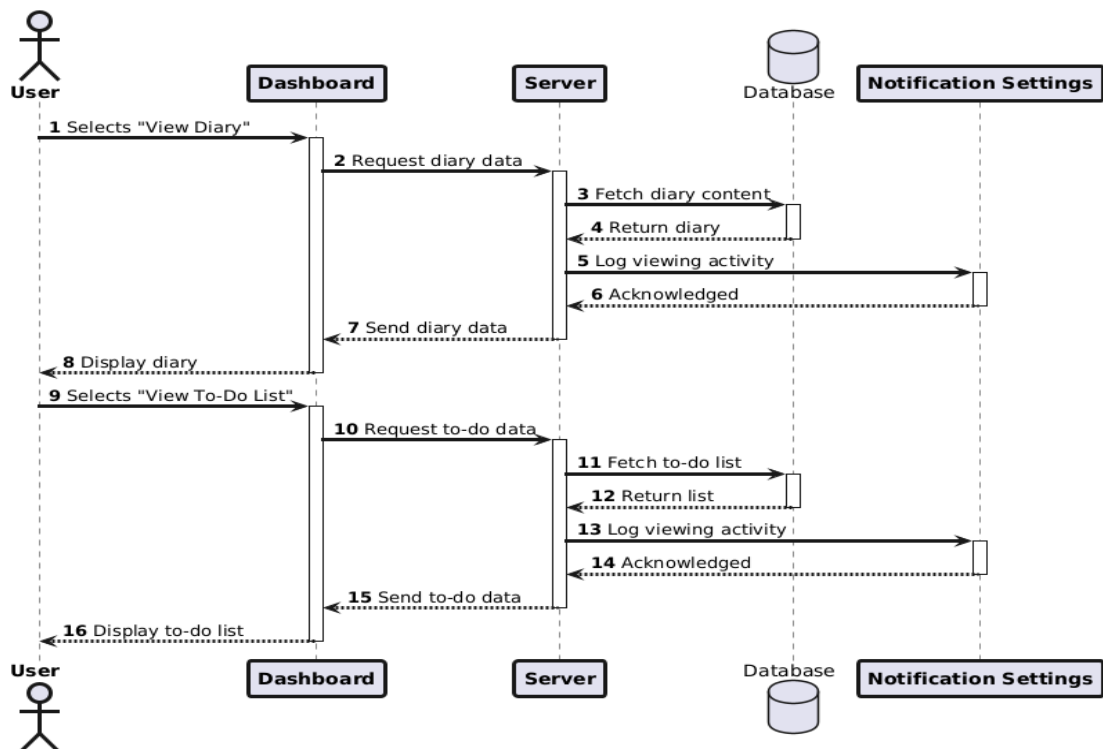
4.5.1 Sequence Diagram – Login Authentication

ii. Signup Authentication



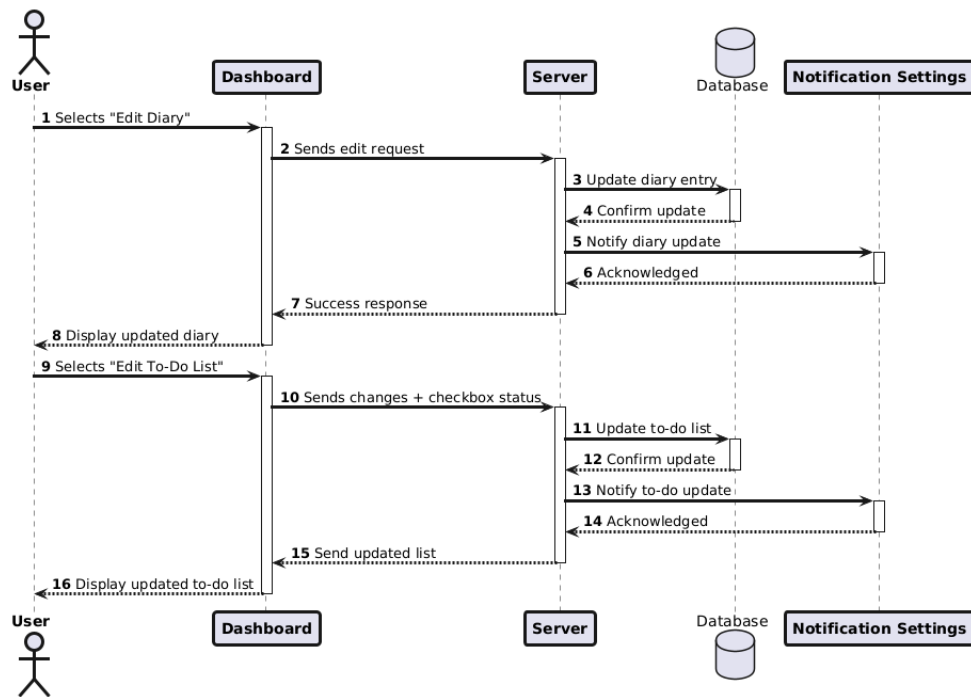
4.5.2 Sequence Diagram – Signup Authentication

iii. View Diary and View Todo List



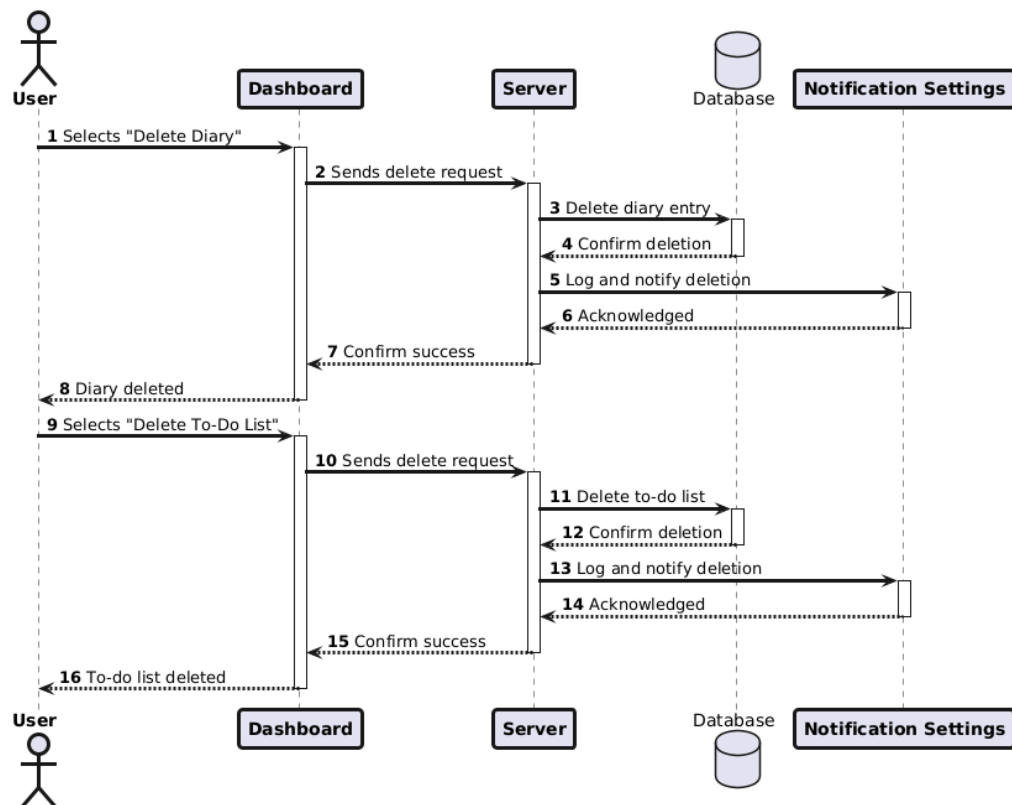
4.5.3 Sequence Diagram – View Diary and ToDo

iv. Edit Diary and Edit Todo List



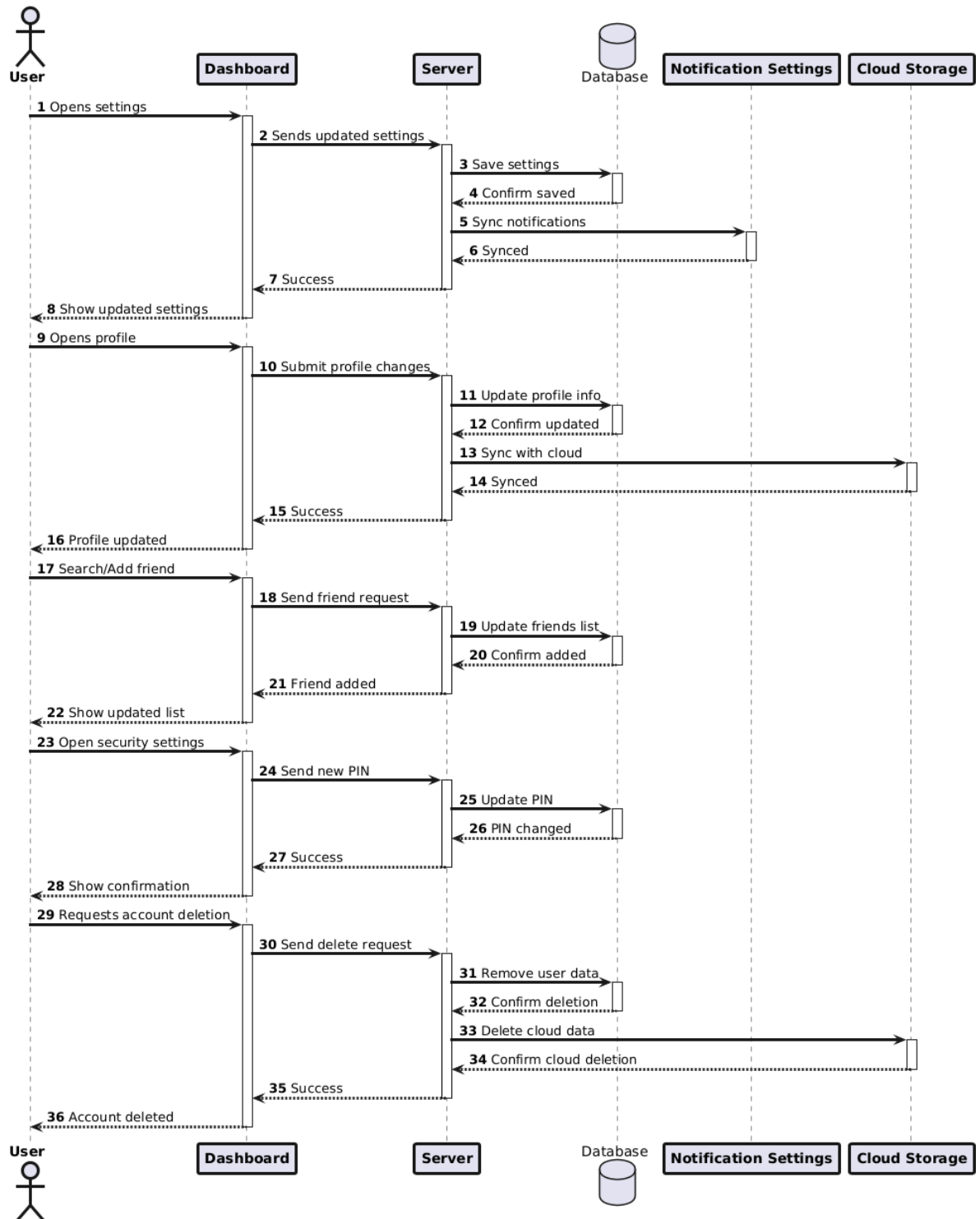
4.5.4 Sequence Diagram – Edit Diary and Edit ToDo

v. Delete Diary and Delete Todo List



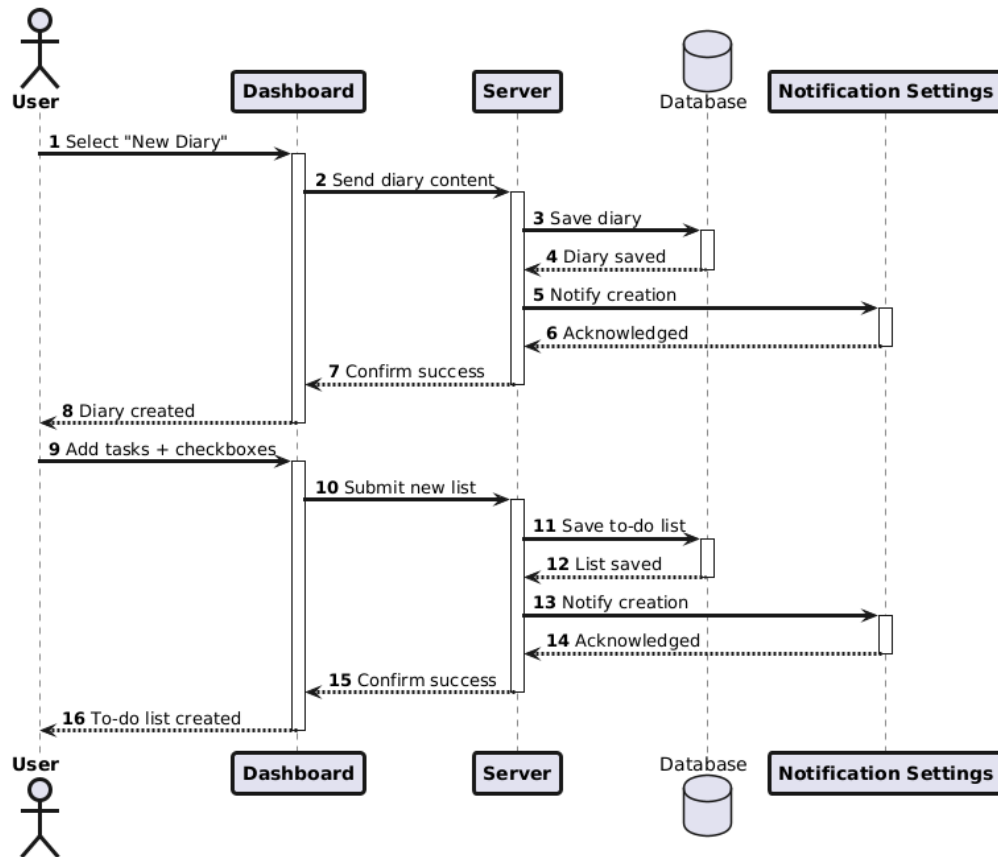
4.5.5 Sequence Diagram – Delete Diary and ToDo

vi. Edit Settings, Profile, Add Friends, and Account Deletion



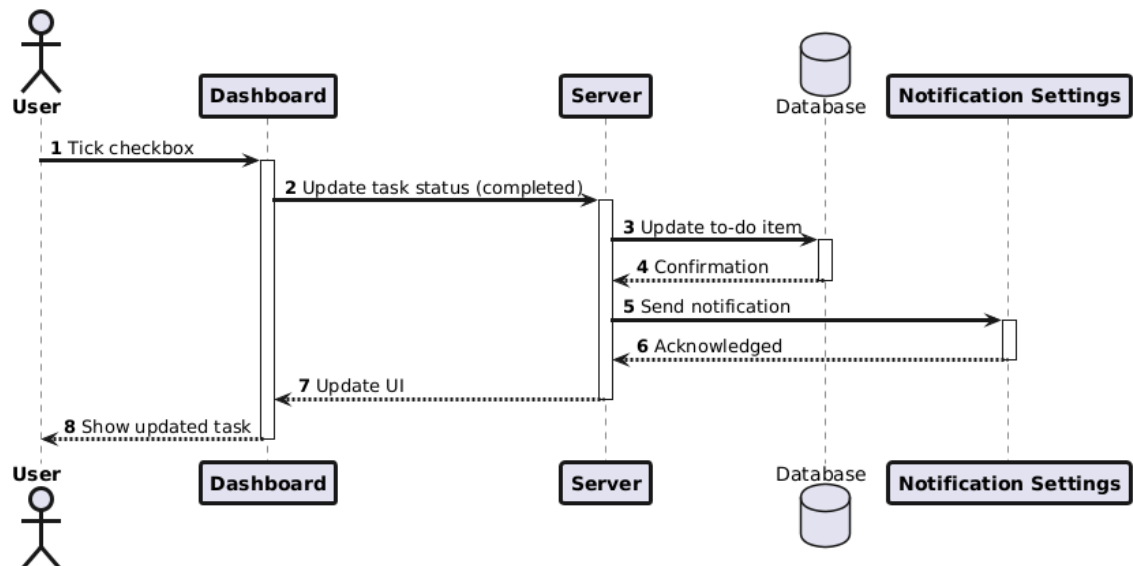
4.5.6 Sequence Diagram - Edit Settings, Profile, Add Friends, and Account Deletion

vii. Create Diary and ToDo



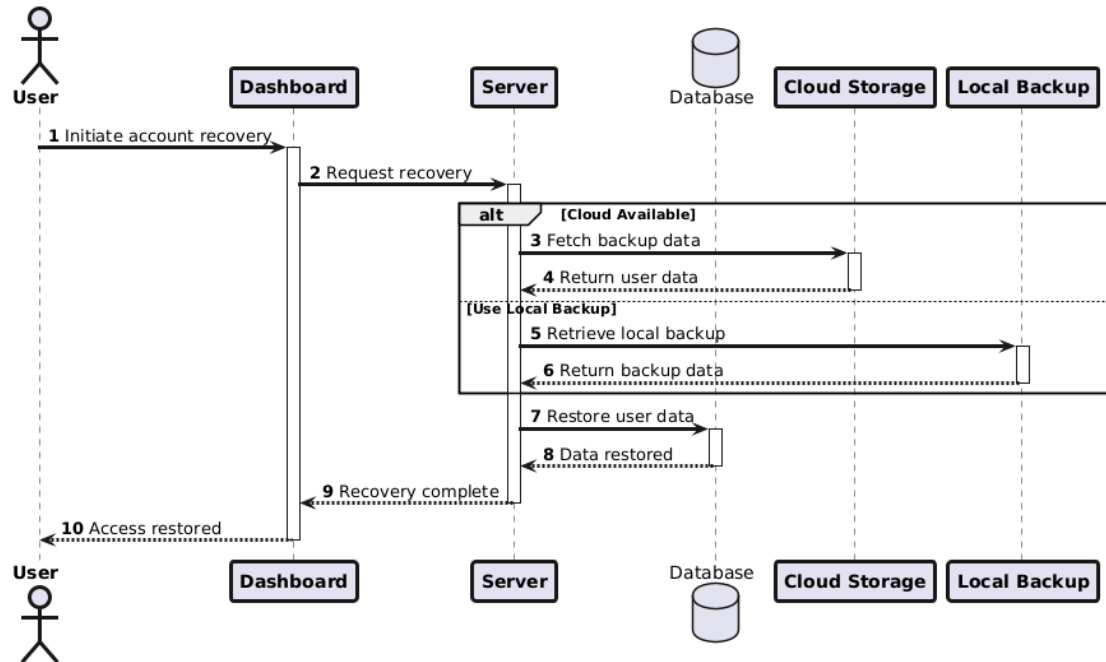
4.5.7 Sequence Diagram – Create Diary and ToDo

viii. Mark ToDo as Complete



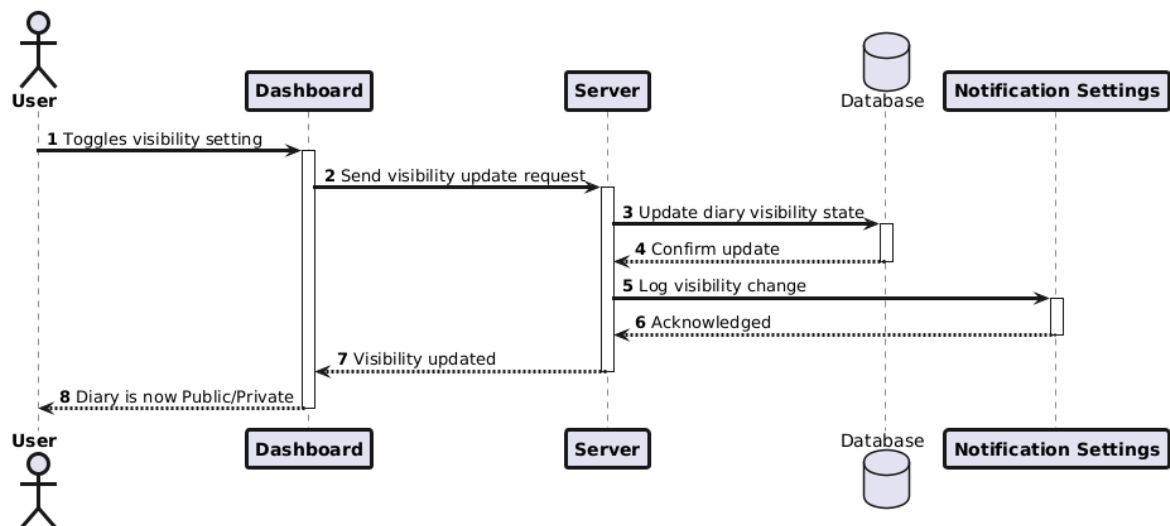
4.5.8 Sequence Diagram – Mark ToDo Complete

ix. Account Recovery



4.5.9 Sequence Diagram – Account Recovery

x. Change Diary Visibility



4.5.10 Sequence Diagram – Public Diary

4.7 Data Dictionary

A Data Dictionary is a collection of names, definitions, and attributes about data elements that are being used or captured in a database, information system, or part of a research project.

We have total of 15 data dictionaries needed in our project to store data and implement the smooth transitions in the data flow.

i. Account

<u>Account</u>				
Element	Data Type	Key	Auto Increment	Description
user_id	integer(8)	PK	Yes	A unique identifier for each user
username	varchar(32)			The unique name chosen by the user to be identified by in the app.
user_email	varchar(64)			The user's email address
user_firstname	varchar(32)			The first name of the user.
user_lastname	varchar(32)			The last name of the user.
user_gender	enum			The gender of the user.
user_dob	DATE			The date of birth of the user.
user_pfp_url	varchar(255)			The URL of the user's profile picture, used to display the profile image.
join_date	timestamp			The date and time when the user created their account.
theme_id	varchar(4)			A reference to the user's chosen theme.

Table 4.7.1: Account Table

ii. Login

<u>Login</u>				
Element	Data Type	Key	Auto Increment	Description
user_id	integer(8)	FK		A unique identifier for each user
username	varchar(32)			The unique name chosen by the user to be identified by in the app.
password	varchar(64)			The user's password for account access.
user_pin	varchar(4)			The PIN set by the user for an additional login method.
pin_enabled	boolean			Indicates whether the PIN login option is enabled (true/false).
login_available	boolean			Specifies if the user is currently allowed to log in.
login_attempy	integer(1)			Tracks the number of failed login attempts(max 3).

Table 4.7.2: Login Table

iii. Diary

<u>Diary</u>				
Element	Data Type	Key	Auto Increment	Description
diary_id	integer(8)	PK	Yes	A unique identifier for each diary entry.
user_id	integer(8)	FK		A unique identifier for each user.
profile_visibility	boolean			Indicates whether the diary entry is public or private.
created_at	timestamp			The date and time when the diary entry was created.
last_modified	timestamp			The date and time when the diary entry was last modified.
content_link	varchar(255)			A link to the stored content of the diary.

Table 4.7.3: Diary Table

iv. Tag

<u>Tag</u>				
Element	Data Type	Key	Auto Increment	Description
tag_id	integer(8)	PK	Yes	A unique identifier for each tag.
tag_name	varchar(24)			The name of the tag used to label diary entries.
tag_description	varchar(64)			A brief description of the tag's purpose or meaning.

Table 4.7.4: Tag Table

v. Tag Entries

<u>Tag Entries</u>				
Element	Data Type	Key	Auto Increment	Description
diary_id	integer(8)	FK		A unique identifier for the diary entry associated with the tag.
tag_id	integer(8)	FK		A unique identifier for the tag.

Table 4.7.5: Tag Entries

vi. Notification

<u>Notification</u>				
Element	Data Type	Key	Auto Increment	Description
noti_id	integer(8)	PK	Yes	A unique identifier for each notification.
noti_type	varchar(24)			The type of notification(reminder, alert).
noti_desc	varchar(100)			A description of the notification.
noti_time	date			The time when the notification is scheduled.
noti_date	date			The date when the notification is scheduled.
noti_frequency	integer(2)			The frequency of the notification.

Table 4.7.6: Notification Table

vii. **Badge**

<u><i>Badge</i></u>				
Element	Data Type	Key	Auto Increment	Description
badge_number	integer(8)	PK	Yes	A unique identifier for each badge.
badge_name	varchar(24)			The name of the badge earned by the user.
Badge_type	varchar(24)			The type or category of the badge.
Badge_link	varchar(255)			A link to the image or content associated with the badge.

Table 4.7.7 Badge Table

viii. User Badge Info

<u><i>User Badge Info</i></u>				
Element	Data Type	Key	Auto Increment	Description
badge_id	integer(8)	PK	Yes	A unique identifier for each badge info record.
badge_number	integer(8)	FK		A unique identifier for the badge associated with the user.
user_id	integer(8)	Fk		A unique identifier for each user.
received_date	timestamp			The date and time when the badge was received.

Table 4.7.8: User Badge Info Table

ix. Backup

<u>Backup</u>				
Element	Data Type	Key	Auto Increment	Description
backup_id	integer(8)	PK	Yes	A unique identifier for each backup record.
user_id	integer(8)	FK		A unique identifier for each user.
backup_date	timestamp			The date and time when the backup was created.
backup_location	varchar(255)			The location where the backup is stored.
backup_status	boolean			Indicates whether the backup(successful or failed).

Table 4.7.9: Backup Table

x. Friend List

<u>Friend List</u>				
Element	Data Type	Key	Auto Increment	Description
friend_number	integer(4)	PK	Yes	A unique identifier for each friend record.
user_id	integer(8)	FK		A unique identifier for each user.
friend_userid	enum			
request_status	boolean			Indicates whether the friend request is pending, accepted, or rejected.

Table 4.7.10: Friend Table

xi. Theme

<u>Theme</u>				
Element	Data Type	Key	Auto Increment	Description
theme_id	integer(8)	PK	Yes	A unique identifier for each theme.
theme_name	varchar(24)			The name of the theme.
theme_desc	varchar(64)			A description of the theme.
format_link	varchar(255)			A link to the format or design file associated with the theme.

Table 4.7.11: Theme Table

xii. Template

<u>Template</u>				
Element	Data Type	Key	Auto Increment	Description
template_id	integer(8)	PK	Yes	A unique identifier for each template.
user_id	integer(8)	FK		A unique identifier for each user.
template_name	varchar(24)			The name of the template.
template_link	varchar(255)			A link to the template content or file.

Table 4.7.12: Template Table

xiii. Reaction

<u><i>Reaction</i></u>				
Element	Data Type	Key	Auto Increment	Description
react_number	integer(8)	PK	Yes	A unique identifier for each reaction record.
react_id	integer(8)	FK		A unique identifier for the reaction type.
user_id	integer(8)	FK		A unique identifier for each user.
dairy_id	integer(8)	FK		A unique identifier for each diary entry.

Table 4.7.13: Reaction Table

xiv. Reaction Type

<u><i>Reaction Type</i></u>				
Element	Data Type	Key	Auto Increment	Description
react_id	integer(8)	PK	Yes	A unique identifier for each reaction type.
react_name	varchar(24)			The name of the reaction.
react_desc	varchar(64)			A description of the reaction.

Table 4.7.14: Reaction Type Table

xv. **Streak**

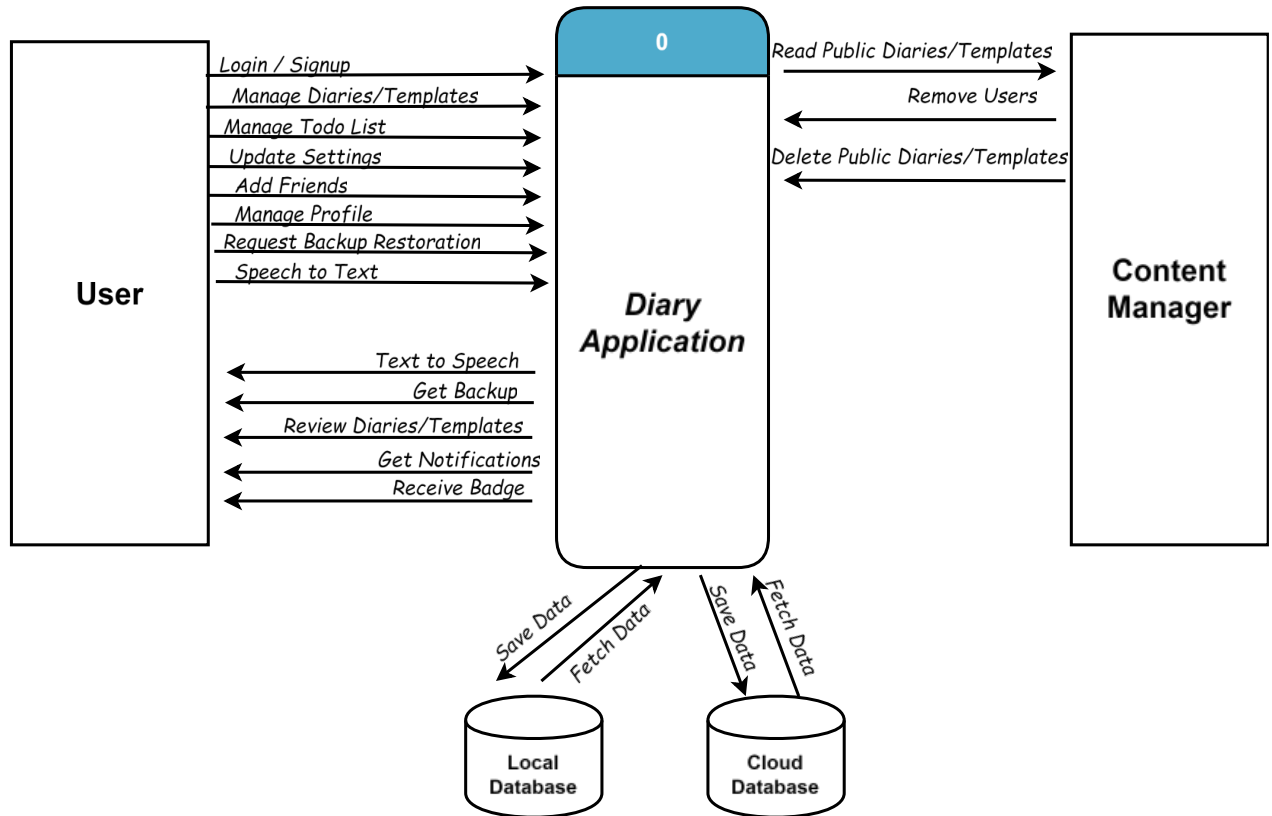
<u><i>Streak</i></u>				
Element	Data Type	Key	Auto Increment	Description
Streak ID	Integer(8)	PK	Yes	A unique identifier for each Streak.
user_id	integer(8)			
Date Started	DateTime			

Table 4.7.15: Streak Table

4.8 Data Flow Diagram

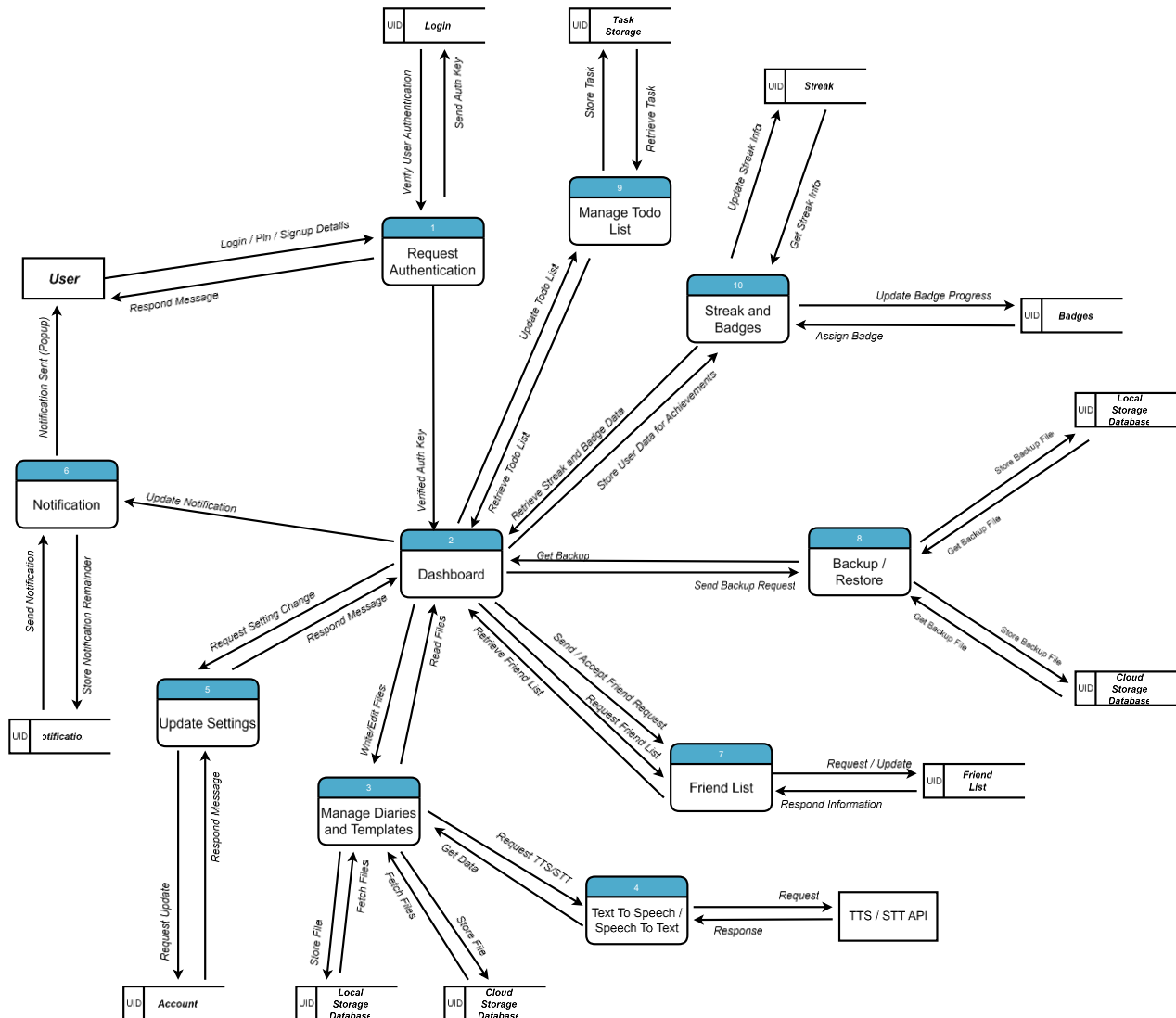
Data Flow Diagram (DFD) represents the flow of data within information systems.

4.8.1 Level 0 DFD



4.8.1 Level 0 DFD

4.8.2 Level 1 DFD

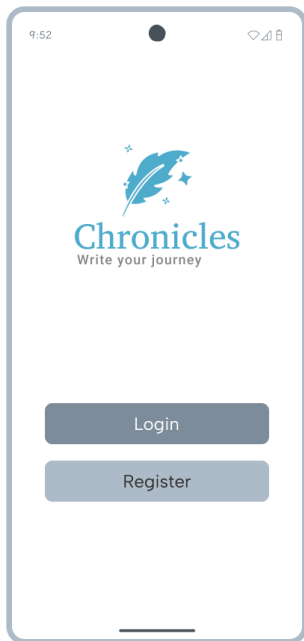


4.8.2 Level 1 DFD

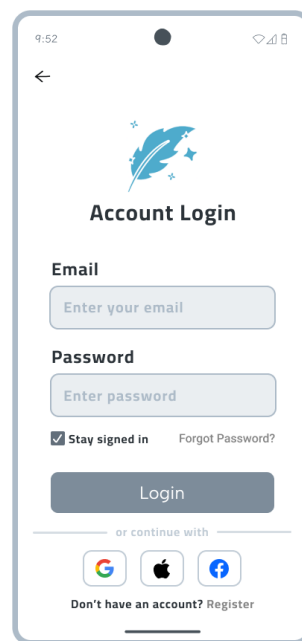
4.9 Wireframes

A wireframe is a rough schematic created in the early stages of digital product design to help visualize and communicate the structure of a product or website.

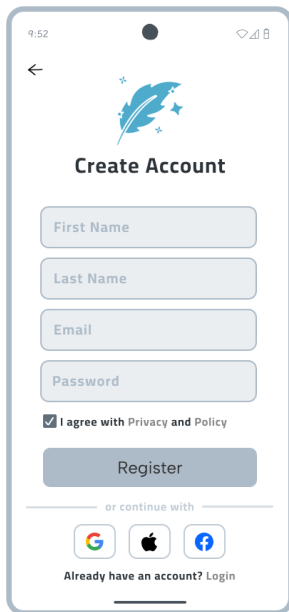
i. Login and Register Wireframes



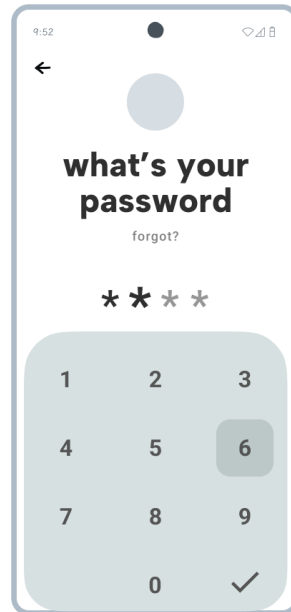
4.9.1 Welcome Screen



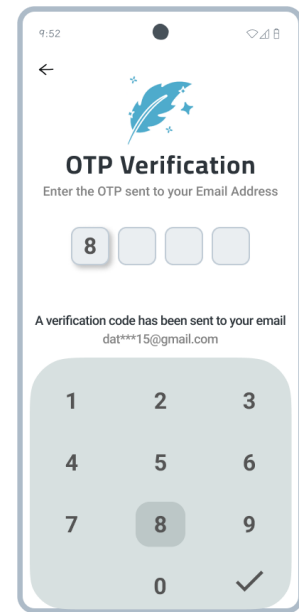
4.9.2 Login Screen



4.9.3 Register Screen

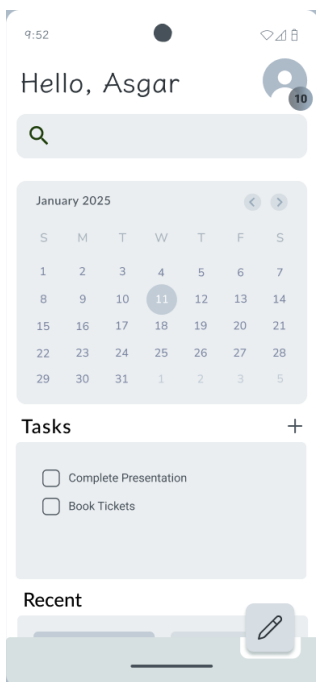


4.9.4 Pin Screen

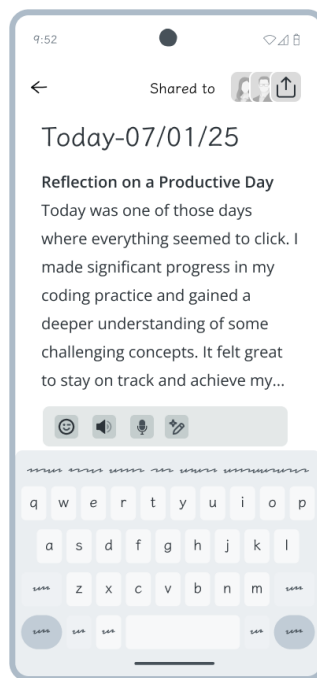


4.9.5 OTP Verification Screen

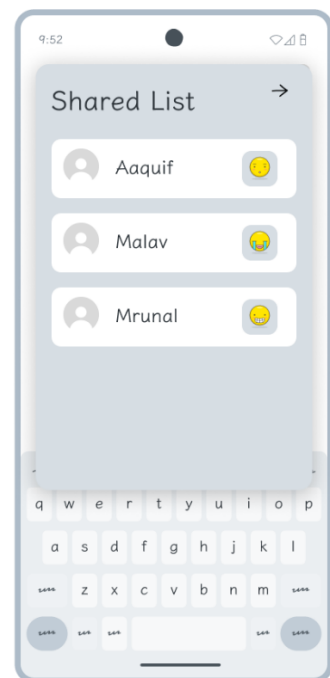
ii. Dashboard, Diary Writing, Sharing Diary, Settings Wireframe



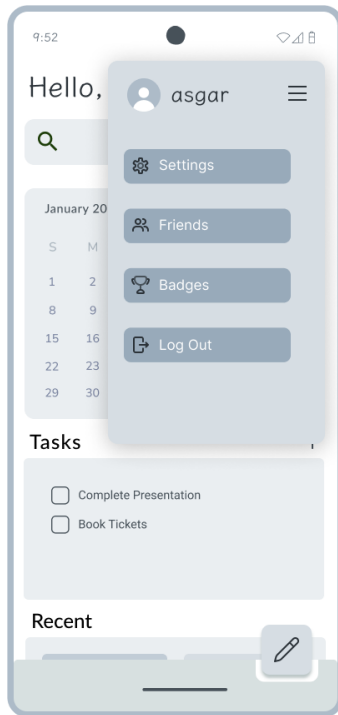
4.9.6 Dashboard Screen



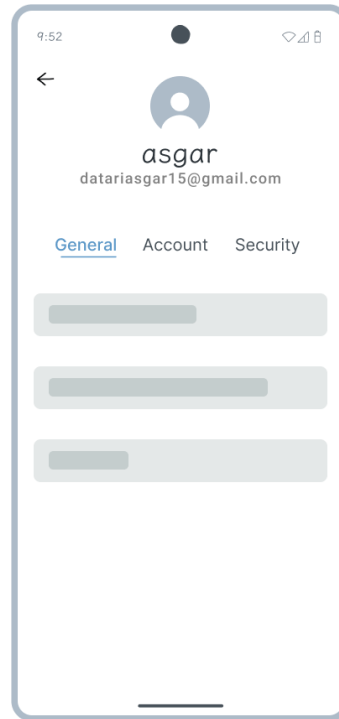
4.9.7 Text Editor Screen



4.9.8 Share With Friends In Text Editor Screen

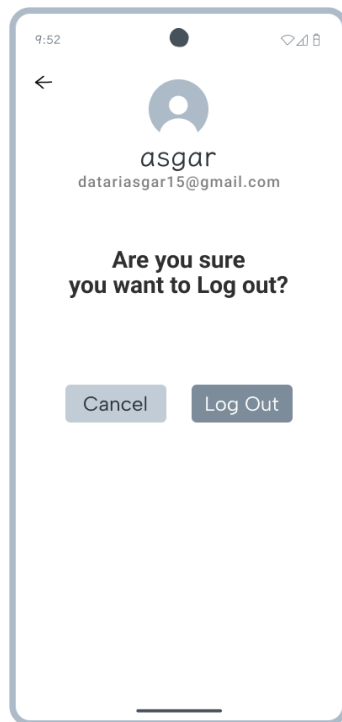


4.9.9 Settings Popup Screen



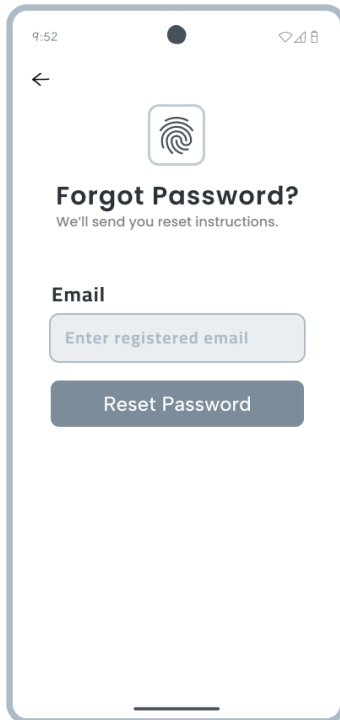
4.9.10 Profile Settings Screen

iii. Logout wireframe

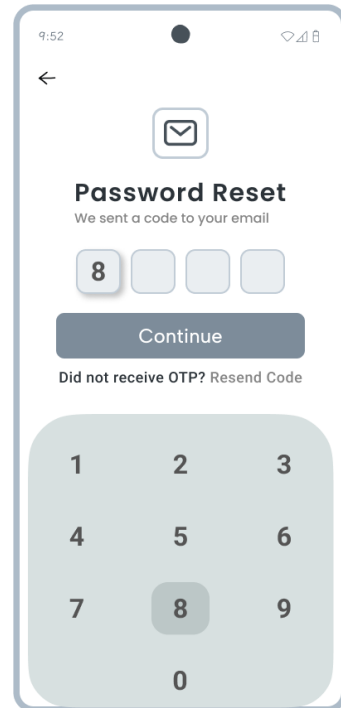


4.9.11 Logout Screen

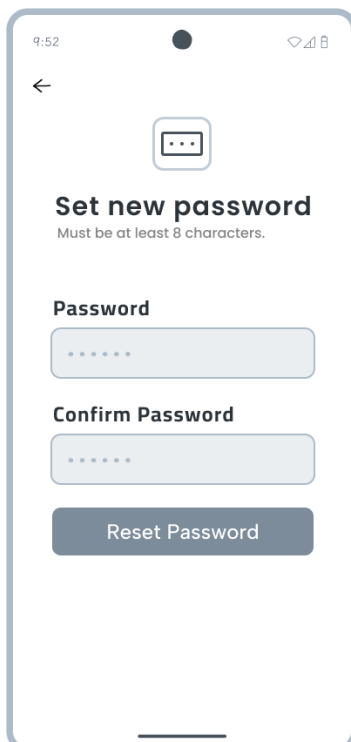
iv. Change Password Wireframe



4.9.12 Forgot Password Screen

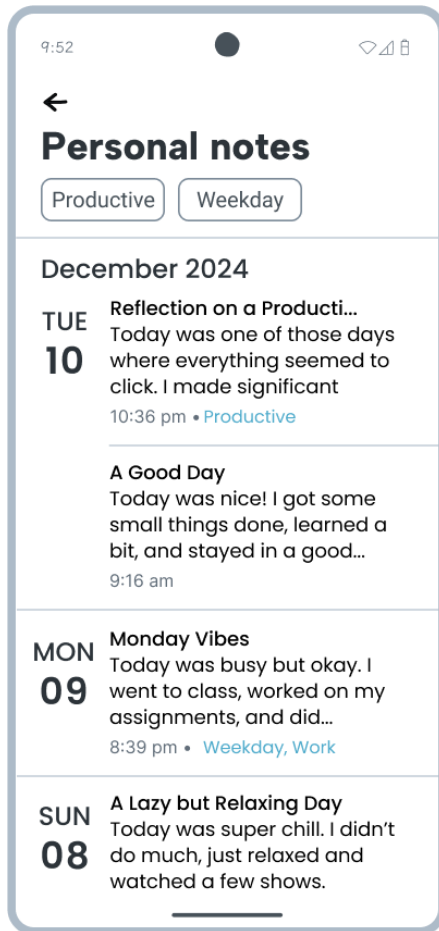


4.9.13 Password Reset Verification Screen



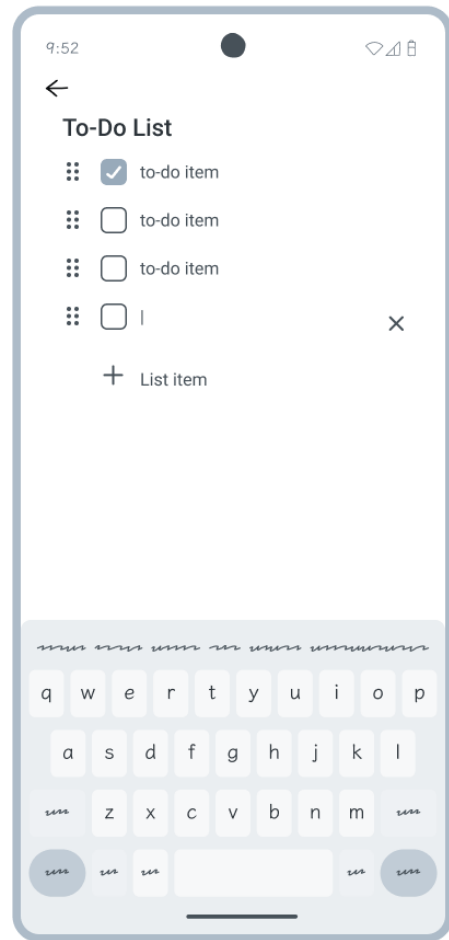
4.9.14 Change Password Screen

v. Personal Notes Wireframe

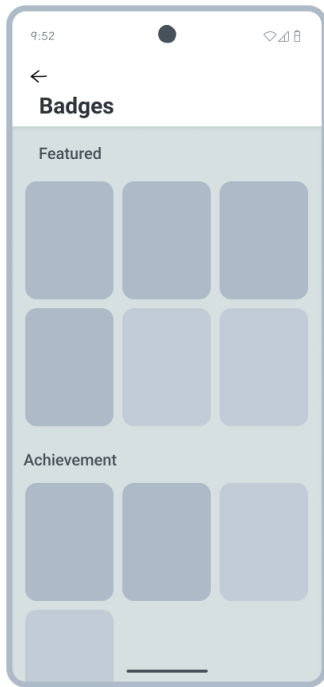


4.9.15 Archive Screen

vi. ToDo List Wireframe



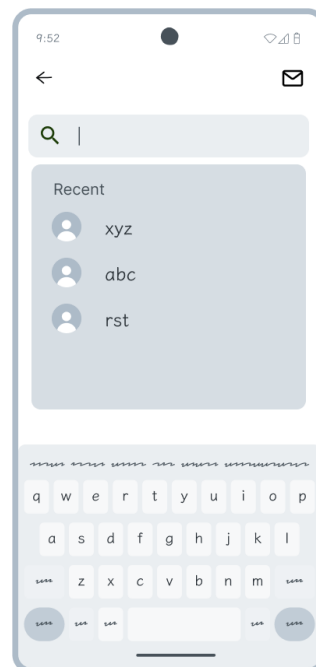
4.9.16 ToDo List

vii. Badges Wireframe

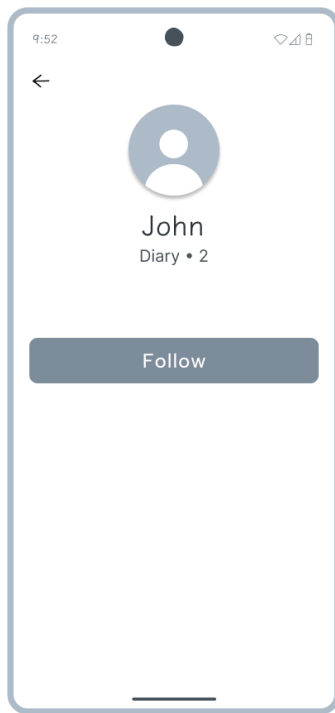
4.9.17 Badge Screen

viii. Search Friends

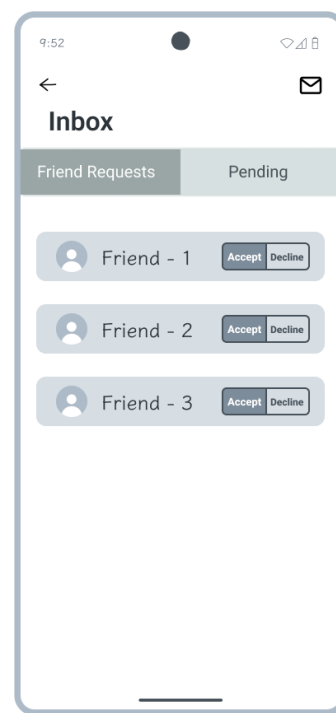
4.9.18 Friend List Screen



4.9.19 Search Friend Screen



4.9.20 Follow Friend Screen



4.9.21 Accept, Pending Friend Inbox Screen

Wireframes were our design layouts, we used to make better UIUX for our application. We changed aspects in wireframes based on user friendliness, and simple to keep UI. Our Project Screenshots are attached in Chapter 8.

Chapter 5: Limitations

5.1 Overview

Area of Limitations is domain where we can face some problems. This domain can include technology stack we use, tools we use, Software dependencies and more.

5.2 Limitations

5.2.1 Device Based Limitations

- i. We may not be able to provide device compatibility for all devices.
- ii. All device has different screen widths, different keyboard and different functionality. Sometimes packages are not available in all application to make the feature work.

5.2.2 Software Based Limitations

- i. Text Summarizer, we using an API for text summarizer, any policy change can affect us.
- ii. Flutter framework is constantly upgrading, so any feature and language update will affect us, like we won't be able to use new features until we update the codebase to existing rules.
- iii. Android and IOS version change do affect applications.

5.2.3 Diary Writing Limitations

- i. Writing diary is tedious and time consuming, even in digital form. We do provide speech to text but even though many people won't be able to make it a habit.

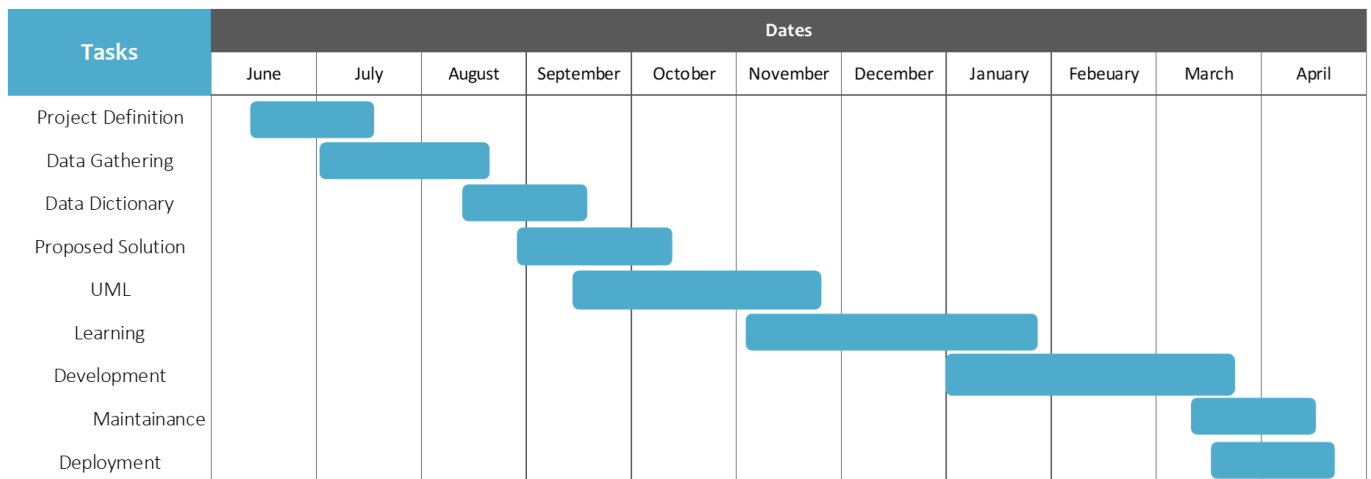
Chapter 6: Timeline

6.1 Overview in Project Timeline

We have broken down the project implementation into 9 phases of development in our timeline. These 9 phases are as follows:

- i. **Project Definition:** In this phase, we define the project's problem statement and provide an overview of the problem and its proposed solution.
- ii. **Data Gathering:** During this phase, we collect data related to the problem statement, explore available alternatives, and identify resources.
- iii. **Proposed Solution:** In this phase, we design the solution on paper, using wireframes and UML diagrams to visualize the overall implementation of the application.
- iv. **Learning:** In this phase, we acquire the necessary skills and knowledge of the tools and technologies required to implement the proposed solution.
- v. **Development:** In this phase, we apply the tools and technologies to implement the application in real-time.
- vi. **Testing:** During this phase, we thoroughly test the software we have developed to ensure its functionality and performance.
- vii. **Deployment:** In this phase, we deploy the software to the intended environment.
- viii. **Maintenance:** After deployment, we handle bug fixes, updates, and patches to ensure the software continues to function optimally.

6.2 Our Projected Timeline



Chapter 7: Tools & Technologies

7.1 Breakdown the development:

This project has different parts where as a group, we allocated different domains to different members.

- i. Frontend Development
- ii. Backend Development
- iii. Database Management
- iv. AI Development
- v. System Design and UI-UX Design

7.2 Tools & Technologies we used:

Front end development	Back end development	Database management	AI development	Designing and UML Development
 Flutter	 Flutter, Dart	 SQLite Supabase, Firebase	 Python, Dart	 draw.io, Figma
Qureshi Aaquif Shah Mrunal	Qureshi Aaquif Shah Mrunal	Qureshi Aaquif Shah Mrunal	Shah Mrunal	Shah Mrunal Datari Asgar Qureshi Aaquif Shah Malav

- i. For **Frontend** Development, we are going to use Flutter Framework. In this part we build the UIUX or User Interface part.
- ii. For **Backend** Development, we are going to use Flutter Framework and Dart Language. Here, we build the core application which is heart of the whole application.
- iii. For **Database** Development, we going to use SQLite, Firebase, and Supabase. In this we use database to store data and fetch it when needed.
- iv. For **System design** and UIUX Designing, we use Figma, and Draw.io. All our chapter 4 Diagrams and Wireframes are built on it.

Chapter 8: Project Screenshots

8.1 Glimpse of Our Application:

As we develop a cross-platform application which has compatibility on both android & iOS. We would be eager to show the development of our project till this date of writing this documentation.

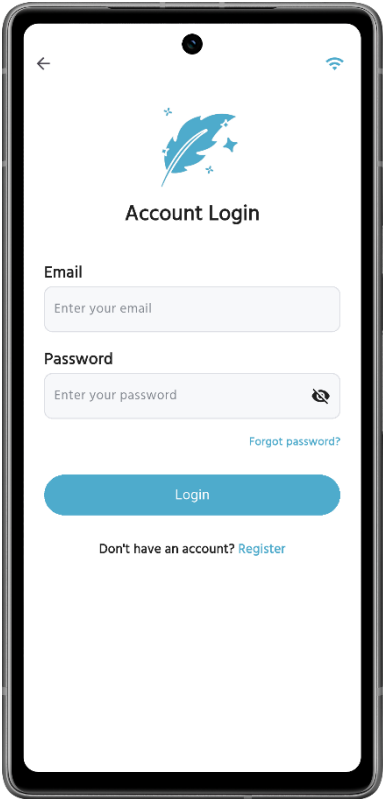
We would divide this chapter into parts based on different components based on our developmental planning.

1. Authentication
2. Dashboard
3. Text Editor & Archive
4. To Do List
5. Settings
6. Friend List

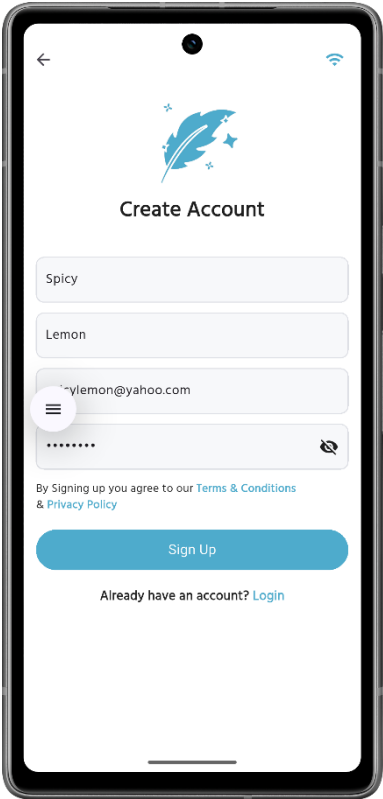
14.1.1. Authentication



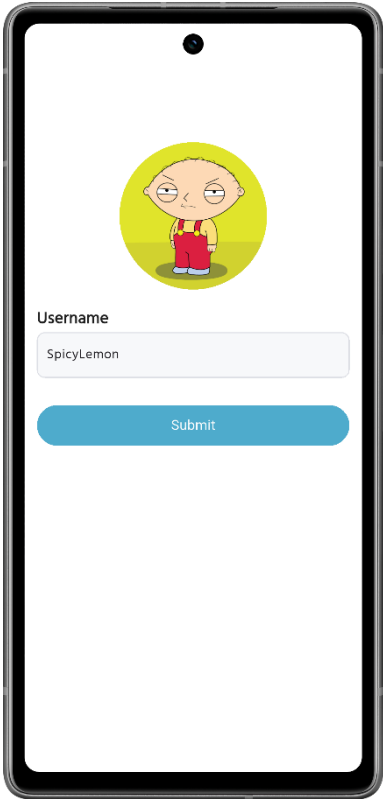
8.1.1.1 Welcome Screen



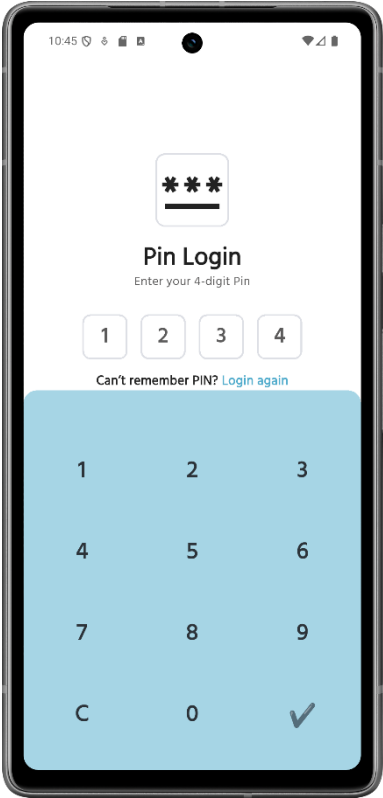
8.1.1.2 Login Screen



8.1.1.3 Register Screen

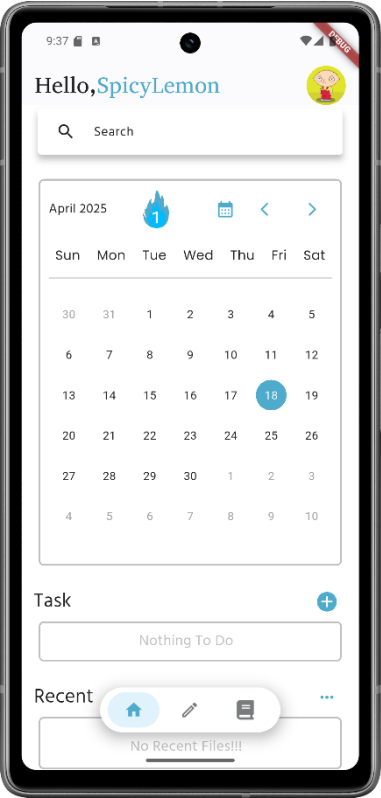


8.1.1.4 Username Enter Screen

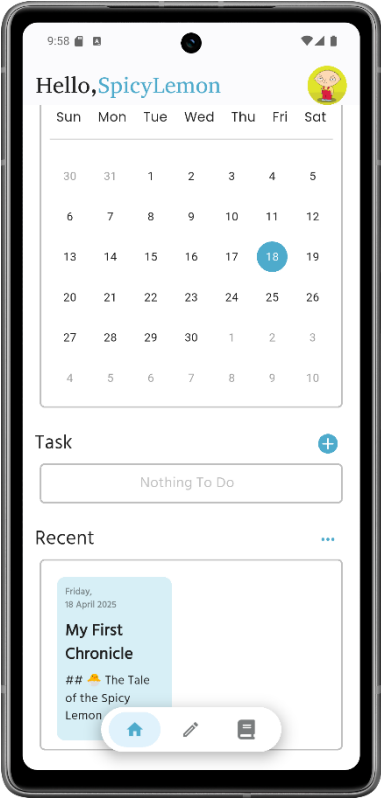


8.1.1.5 Pin Enter Screen

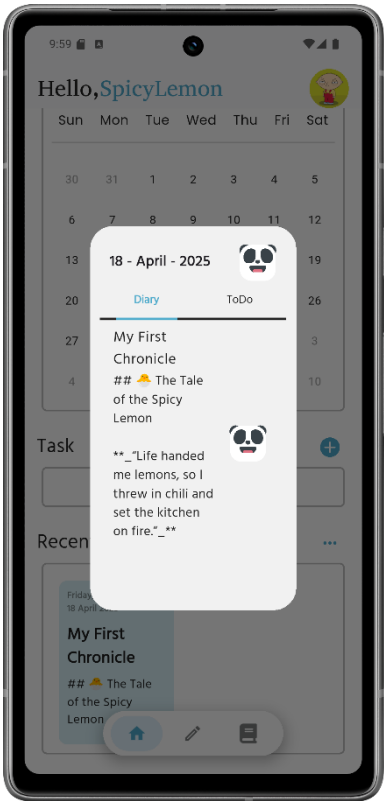
14.1.2. Dashboard



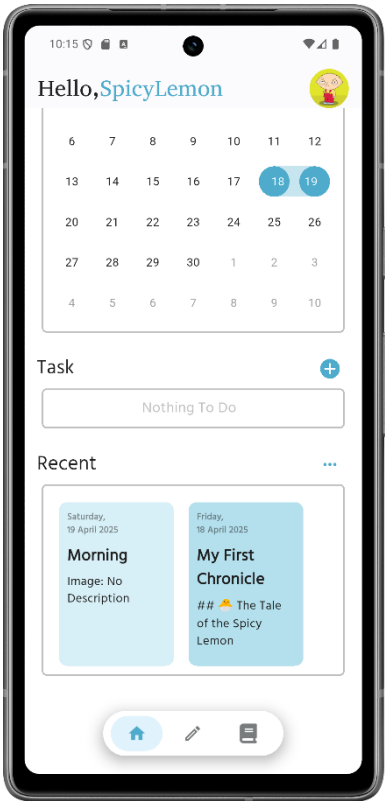
8.1.2.1 Empty Dashboard Screen



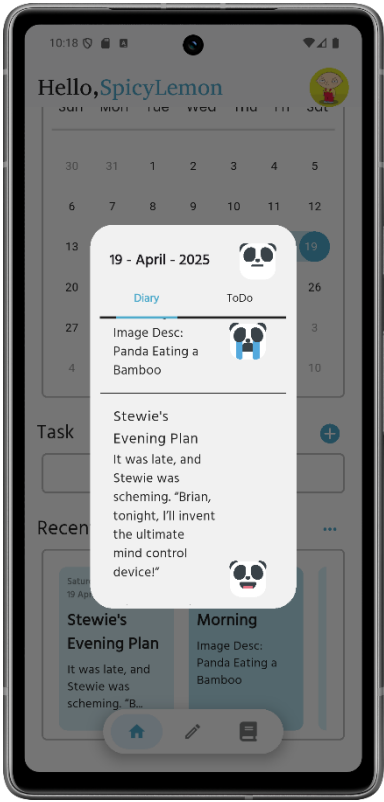
8.1.2.2 First Diary Entry Dashboard Screen



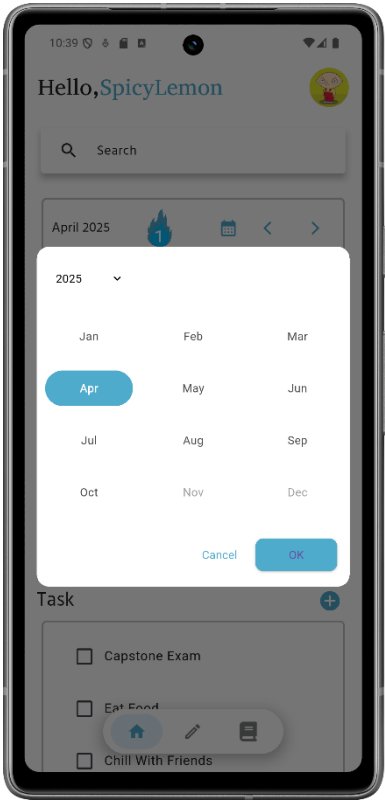
8.1.2.3 View Particular Date Diaries



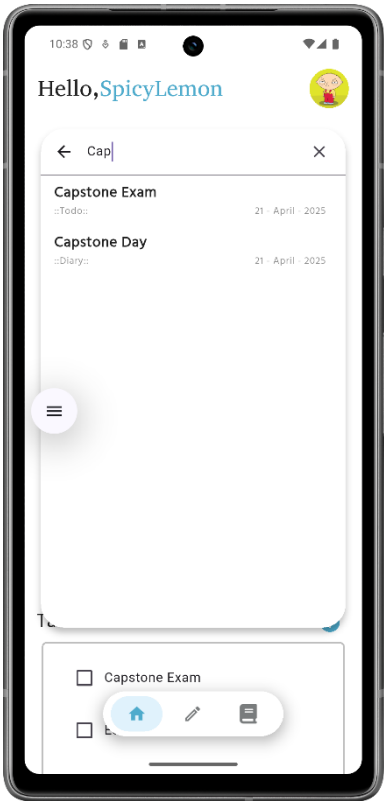
8.1.2.4 Diary Entry Dashboard Screen



8.1.2.3 View Particular Date Diaries

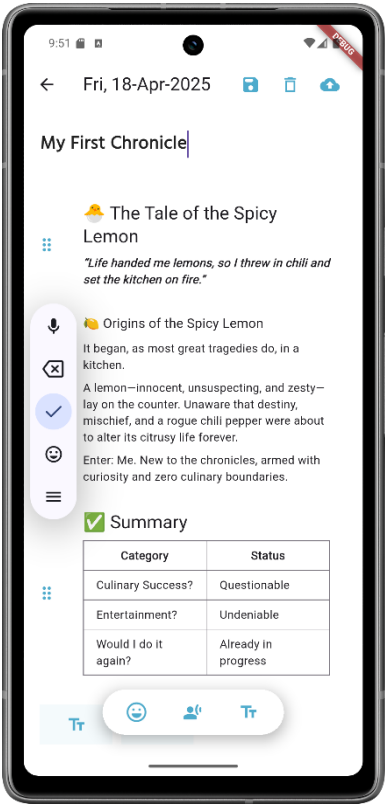


8.1.2.4 Diary Entry Dashboard Screen

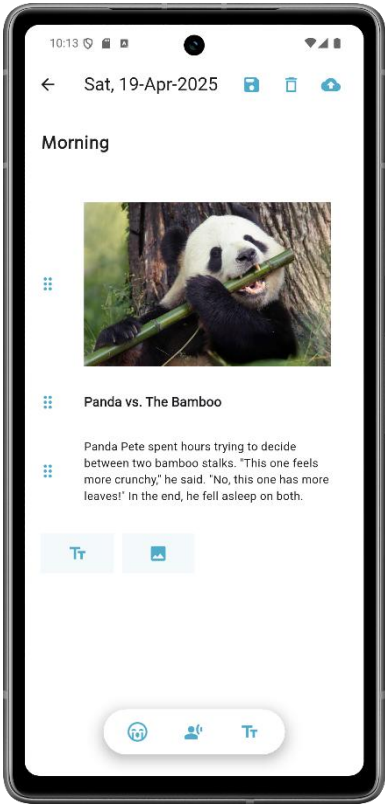


8.1.2.5 Search Bar

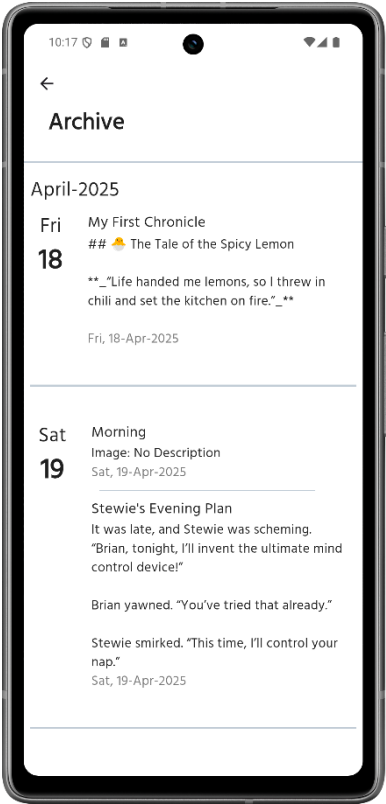
14.1.3. Text Editor & Archive



8.1.3.1 Text Editor Look 1.0

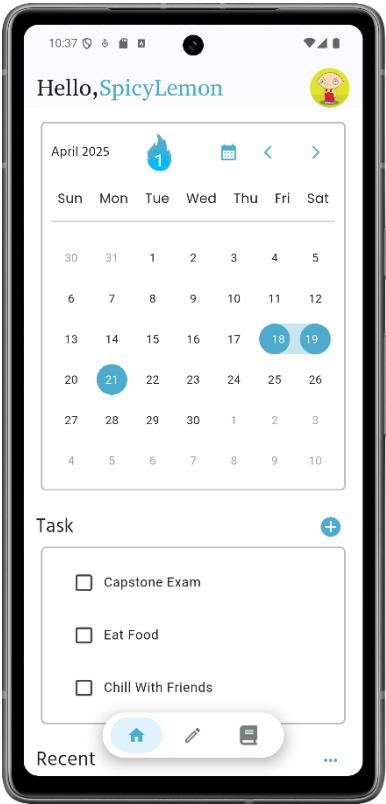


8.1.3.2 Text Editor Look 2.0

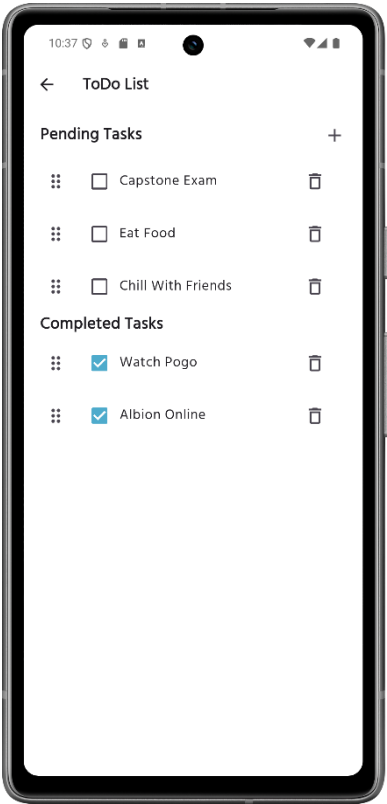


8.1.3.3 Archive Look 1.0

14.1.4. To Do List

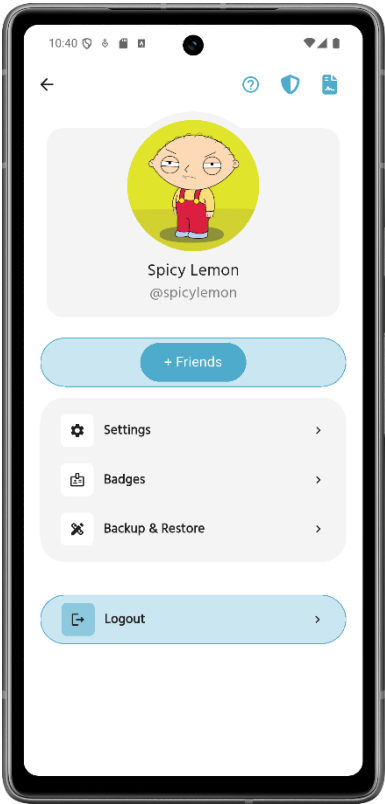


8.1.4.1 ToDo Dashboard View

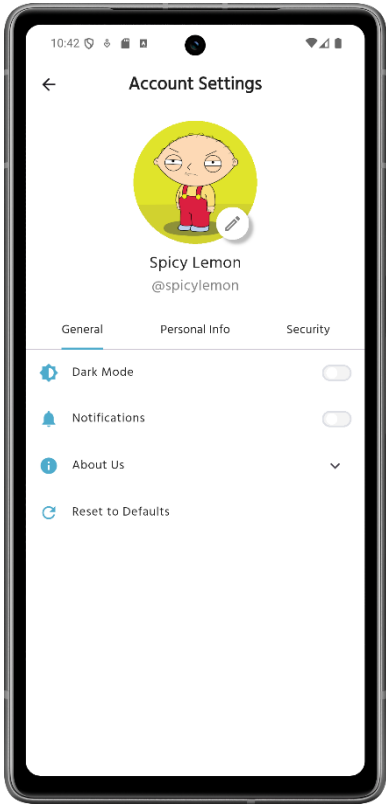


8.1.4.2 ToDo List

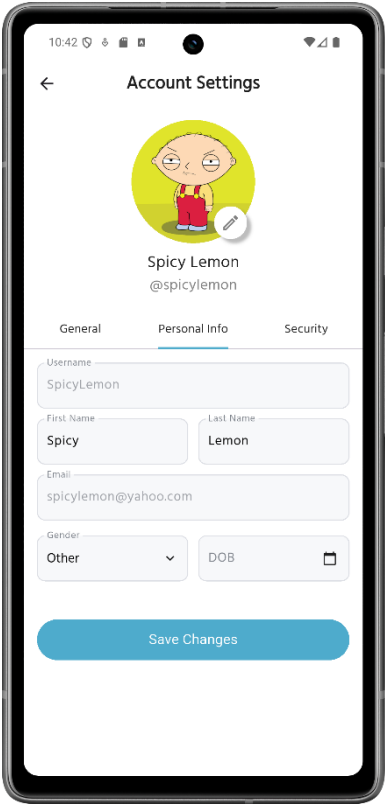
14.1.5. Settings



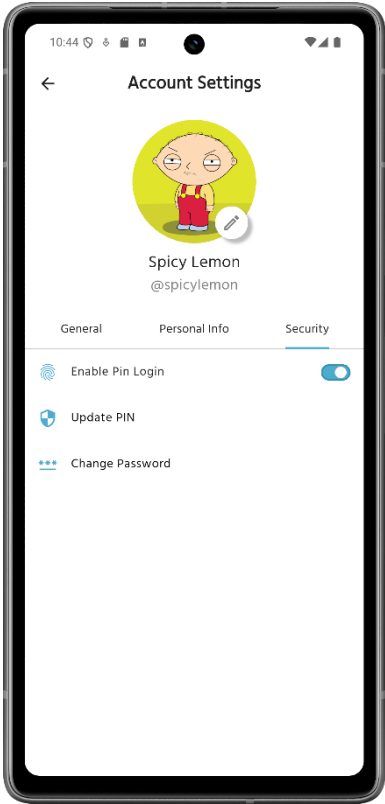
8.1.5.1 Settings Screen



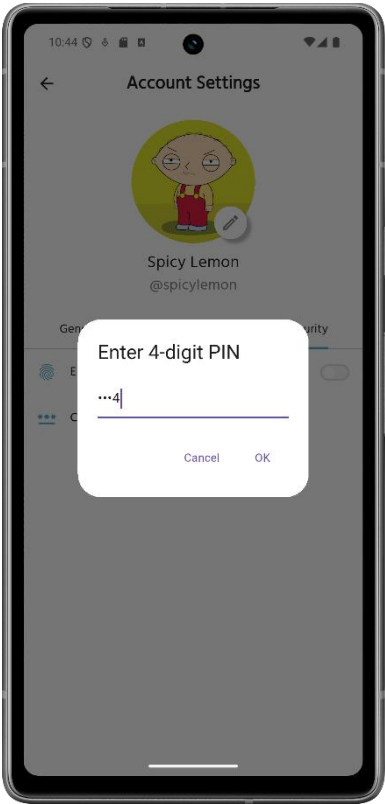
8.1.5.2 General Settings Screen



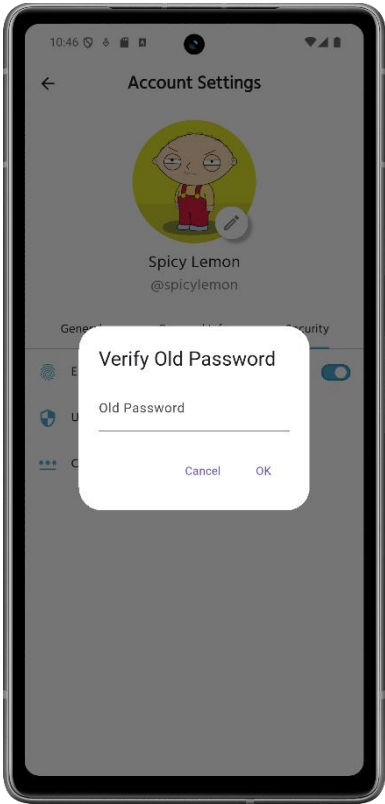
8.1.5.3 Personal Settings Screen



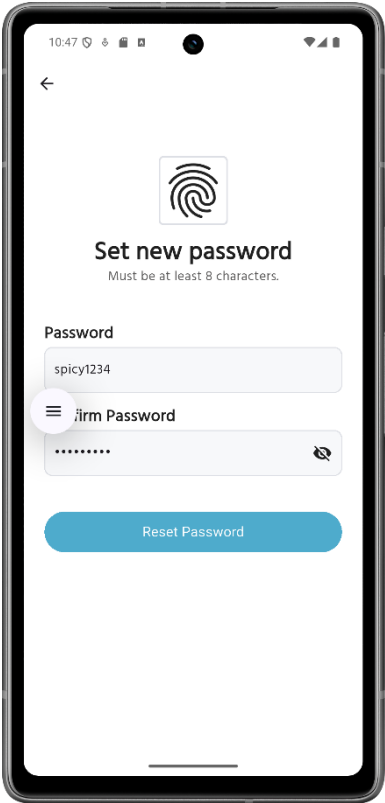
8.1.5.4 Security Settings Screen



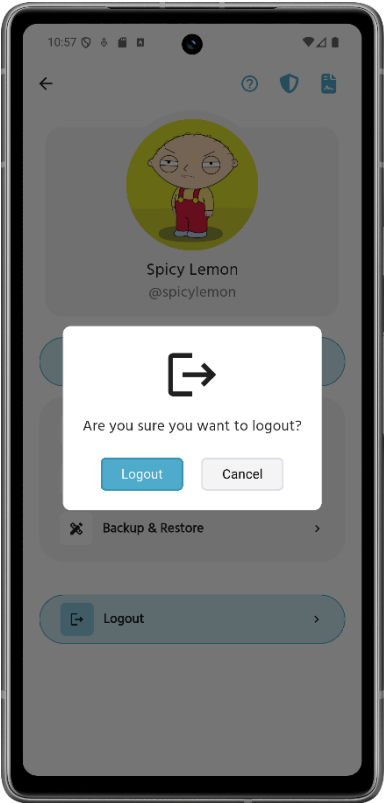
8.1.5.5 Add Pin Security



8.1.5.6 Verify Old Password

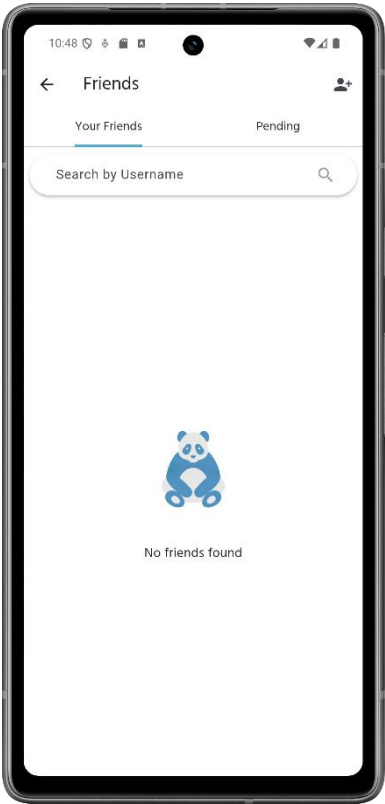


8.1.5.7 Change Password

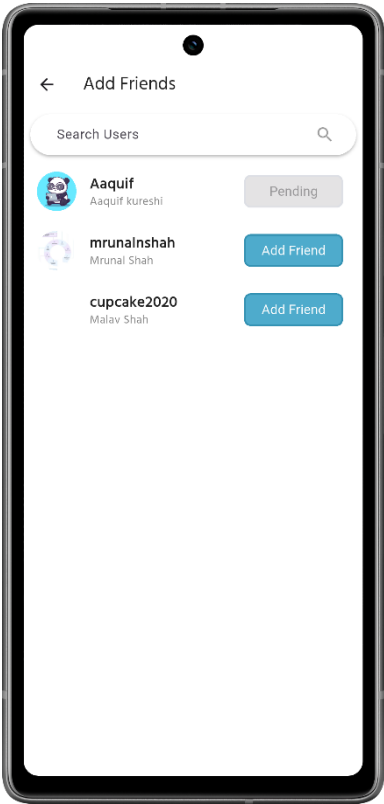


8.1.5.8 LogOut PopUp

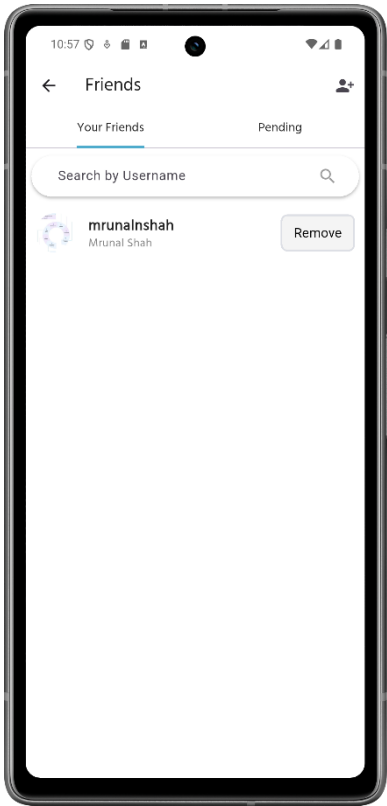
14.1.6. Friends List



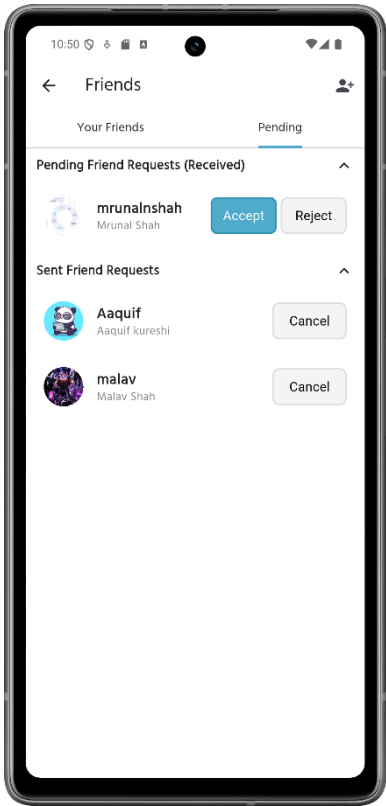
8.1.6.1 Friends Screen



8.1.6.2 Add Friends



8.1.6.3 Friends Screen With Friend Added



8.1.6.4 Accept, Approve Friends

Chapter 9: Conclusion & Future Scope

9.1 Conclusion:

The "Chronicles – Write Your Journey" diary writing app aims to address the growing issue of emotional disconnect and mental stress by encouraging individuals to engage in regular self-reflection. Through an easy-to-use, cross-platform solution, the app allows users to digitize their thoughts, emotions, and life events in a way that is secure, private, and accessible. By offering features such as speech-to-text, habit tracking, templates, and a public diary option, the app enhances self-awareness, promotes personal growth, and fosters a supportive community for users to share their experiences. The integration of artificial intelligence, like text summarization and AI-driven reactions to entries, further enriches the journaling experience, making it not only a tool for reflection but also a method of engaging with one's emotional journey.

This app's approach is grounded in the belief that small daily habits, such as writing a diary, can create significant long-term improvements in mental health and overall well-being. Chronicles provides a flexible and enjoyable platform that encourages regular writing, self-reflection, and personal introspection. The combination of privacy, customizable features, and emotional tracking offers users a comprehensive way to navigate their thoughts and feelings in a digitally connected world.

9.2 Future Scope:

The potential for expanding the Chronicles app is vast. In the future, we can integrate advanced features such as:

- i. **AI Personalization:** Leveraging machine learning to analyze user entries over time and offer personalized suggestions for improving mental well-being, establishing routines, or even predicting emotional trends.
- ii. **Mental Health Integration:** Partnering with mental health professionals to incorporate mood tracking tools and journaling prompts designed to assist individuals in managing stress, anxiety, and depression.
- iii. **Collaborative Features:** Expanding the social element by adding group journaling or allowing shared entries where users can reflect on common experiences, which would help foster a sense of community and mutual support.
- iv. **Multimedia Support:** Further enhancing the ability to add not only text but also images, audio, and even video entries to create a more immersive journaling experience.
- v. **Augmented Reality (AR):** Introducing AR features that allow users to visually interact with past diary entries, bringing memories and reflections to life in a dynamic and engaging way.
- vi. **Integration with Wearable Devices:** Collaborating with wearable technology to gather physiological data, such as heart rate or sleep patterns, and use this data to provide insights on how these factors relate to the user's emotional state.

As we continue to refine and develop Chronicles, the ultimate goal is to create a comprehensive tool for self-care that becomes an integral part of users' lives, offering them not just a way to record their journey, but also to understand and improve it in meaningful ways.

References

1. <https://www.annefrank.org/en/timeline/74/after-an-appeal-from-a-dutch-minister-anne-wants-to-publish-her-diary/>
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